

MENTORA ACADEMY

Cambridge International AS Level Biology (9700)

Personalised Mastery Planner

Student: HAMNA

Target Exam: Cambridge M/J 2027

Date Range: 25 September 2026 — 15 February 2027

Total: 21 Weeks · 144 Study Days · 11 Topics · 550 Practice Questions

WEEK 1: 25 Sep 2026 – 01 Oct 2026

Topic: ENZYMES

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

DAY 1 — Friday, 25 Sep 2026

Topic 3 — ENZYMES (9700 Syllabus 3.1)

Today's Sub-topic: 3.1 Enzyme action

Learning Objective: Explain how enzymes lower activation energy and distinguish lock-and-key from induced-fit models.

What to Learn:

- (1) Activation energy: energy barrier reactants must overcome; enzymes provide alternative pathway of lower E_a .
- (2) Enzymes are globular proteins; active site is specific in shape AND charge — complementary to substrate.
- (3) Lock-and-key model: rigid active site, substrate fits precisely without conformational change.
- (4) Induced-fit model: flexible active site; substrate binding causes conformational change that OPTIMALLY POSITIONS CATALYTIC RESIDUES.
- (5) Evidence for induced-fit: X-ray crystallography of hexokinase shows closure around glucose.
- (6) Enzyme-substrate complex transition state stabilised products released; enzyme unchanged.

How to Outstand:

- EXAMINER TRAP: Never say enzymes 'destroy' or 'remove' activation energy — they LOWER it via an alternative reaction pathway.
- Cambridge awards marks for complementary shape AND CHARGE — not just shape.
- For induced-fit: state conformational change OPTIMALLY POSITIONS CATALYTIC RESIDUES — not just 'it changes shape'.
- Distinguish ESC (enzyme-substrate complex) from transition state — these are different.
- 'Describe' lock-and-key = rigid. 'Explain' induced-fit = WHY the change increases catalysis.

Resources: Q1–Q4 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 2 — Saturday, 26 Sep 2026

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 3 — Sunday, 27 Sep 2026

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 4 — Monday, 28 Sep 2026

Topic 3 — ENZYMES (9700 Syllabus 3.2a)

Today's Sub-topic: 3.2a Substrate concentration & K_m

Learning Objective: Apply Michaelis-Menten kinetics and define K_m as an inverse measure of enzyme-substrate affinity.

What to Learn:

- (1) Low $[S]$: rate proportional to $[S]$ = first-order kinetics.
- (2) High $[S]$: rate plateaus at V_{max} = zero-order kinetics; all active sites occupied simultaneously.
- (3) $K_m = [S]$ at which rate = $\frac{1}{2}V_{max}$. Low K_m = high affinity. High K_m = low affinity.
- (4) Calculate initial rate by drawing tangent at $t=0$ on progress curve; gradient = rate.
- (5) Read K_m from graph: draw horizontal at $\frac{1}{2}V_{max}$ read x-axis.
- (6) V_{max} is an asymptote — never actually reached at finite $[S]$.

How to Outstand:

- EXAMINER TRAP: K_m is NOT the $[S]$ at saturation — it is specifically the $[S]$ at HALF V_{max} .
- Always include UNITS of K_m : mol dm^{-3} .
- V_{max} plateau = all active sites occupied — NOT 'no more substrate'.
- Calculate initial rate using tangent method — state you draw the tangent.
- A* link: K_m approximates K_d of ESC when product formation is rate-limiting.

Resources: Q5–Q9 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 5 — Tuesday, 29 Sep 2026

Topic 3 — ENZYMES (9700 Syllabus 3.2b)

Today's Sub-topic: 3.2b Temperature and pH

Learning Objective: Explain at molecular level how temperature and pH alter enzyme activity through bond disruption.

What to Learn:

- (1) Temperature below optimum: increasing KE more frequent collisions rate rises. $Q_{10} \approx 2$.
- (2) Temperature above optimum: excess thermal energy disrupts hydrogen bonds and ionic bonds maintaining tertiary structure active site shape changes substrate cannot fit denaturation (IRREVERSIBLE).
- (3) pH effect: H/OH alter ionisation state of R-groups (amino acid side chains) lining active site charge and shape change substrate no longer complementary.
- (4) Extreme pH: irreversible denaturation.
- (5) Different enzymes have different pH optima: pepsin pH 2, salivary amylase pH 7, trypsin pH 8.

How to Outstand:

- EXAMINER TRAP: Do NOT say enzyme is 'killed' by heat — say HIGH TEMPERATURE DENATURES the enzyme by BREAKING HYDROGEN BONDS and IONIC BONDS maintaining the tertiary structure.
- Link temperature specifically to BOND TYPES: H-bonds and ionic bonds disrupted — NOT covalent bonds.
- pH: IONISATION STATE OF R-GROUPS changes — state this phrase.
- Denaturation = irreversible; inhibition = reversible — distinguish these.
- $Q_{10} = \text{rate}(T+10^\circ\text{C})/\text{rate}(T^\circ\text{C}) \approx 2$ — know formula for calculation questions.

Resources: Q10–Q14 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 6 — Wednesday, 30 Sep 2026

Topic 3 — ENZYMES (9700 Syllabus 3.2c)

Today's Sub-topic: 3.2c Inhibitors

Learning Objective: Distinguish competitive and non-competitive inhibition kinetically using K_m and V_{max} data.

What to Learn:

- (1) Competitive inhibitor: structurally similar to substrate; binds ACTIVE SITE reversibly; blocks substrate; overcome by excess $[S]$. K_m INCREASES; V_{max} UNCHANGED.
- (2) Non-competitive: binds ALLOSTERIC SITE; conformational change distorts active site; cannot be overcome. V_{max} DECREASES; K_m UNCHANGED.
- (3) Examples: malonate vs succinate dehydrogenase (competitive); heavy metal Hg^2 on SH groups (irreversible non-competitive).
- (4) Allosteric end-product feedback inhibition (e.g. ATP inhibiting phosphofructokinase).
- (5) Lineweaver-Burk plot: competitive shifts x-intercept ($-1/K_m$ changes); non-competitive shifts y-intercept ($1/V_{max}$ changes).

How to Outstand:

- EXAMINER TRAP: Mnemonic for graphs: Competitive = K_m , V_{max} same. Non-competitive = V_{max} , K_m same. Candidates frequently swap these.
- Competitive inhibitor is STRUCTURALLY SIMILAR (complementary) to substrate — not identical.
- Allosteric site is SEPARATE and DISTINCT from active site — state this.
- Heavy metal ions form COVALENT bonds with SH groups IRREVERSIBLE (cannot be diluted away).
- Lineweaver-Burk: know x-intercept = $-1/K_m$, y-intercept = $1/V_{max}$.

Resources: Q15–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 7 — Thursday, 01 Oct 2026

Topic 3 — ENZYMES (9700 Syllabus 3.3)

Today's Sub-topic: 3.3 Enzyme technology

Learning Objective: Explain immobilisation methods, advantages, and industrial/biosensor applications of immobilised enzymes.

What to Learn:

- (1) Immobilisation methods: entrapment in calcium alginate gel beads; covalent bonding to inert support; adsorption to surface; cross-linking with glutaraldehyde.
- (2) Advantages: REUSABLE (enzyme recovered after reaction); PURE PRODUCT (enzyme-free product); THERMOSTABLE (fixed conformation); continuous-flow processing; reduced cost per unit.
- (3) Industrial examples: lactase in alginate beads lactose-free milk; glucose isomerase fructose syrup production.
- (4) Biosensor: enzyme immobilised on electrode; reaction generates electrical signal [substrate]; e.g. glucose oxidase for diabetic blood glucose monitoring.
- (5) Alginate bead protocol: mix enzyme with sodium alginate drip into CaCl₂ gel beads form.

How to Outstand:

- EXAMINER TRAP: Immobilised enzymes are NOT faster — they are REUSABLE and give PURE PRODUCT. Never claim increased speed as an advantage.
- Cambridge mark schemes: state exactly — reusable, pure enzyme-free product, more thermostable, continuous-flow possible.
- Specify the method of immobilisation: entrapment (most common in Cambridge questions) vs covalent bonding.
- Biosensor chain: substrate → enzyme → product → electrical signal → readout. Know the sequence.
- Alginate bead formation: sodium alginate + enzyme dripped into calcium chloride Ca²⁺ cross-links alginate polymer beads trap enzyme.

Resources: Q21–Q25 Section B, Q41–Q45 Section C

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 2: 02 Oct 2026 – 08 Oct 2026

Topic: ENZYMES

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

DAY 8 — Friday, 02 Oct 2026

Topic 3 — ENZYMES (9700 Syllabus 3.1)

Today's Sub-topic: 3.1 Enzyme action

Learning Objective: Explain how enzymes lower activation energy and distinguish lock-and-key from induced-fit models.

What to Learn:

- (1) Activation energy: energy barrier reactants must overcome; enzymes provide alternative pathway of lower E_a .
- (2) Enzymes are globular proteins; active site is specific in shape AND charge — complementary to substrate.
- (3) Lock-and-key model: rigid active site, substrate fits precisely without conformational change.
- (4) Induced-fit model: flexible active site; substrate binding causes conformational change that OPTIMALLY POSITIONS CATALYTIC RESIDUES.
- (5) Evidence for induced-fit: X-ray crystallography of hexokinase shows closure around glucose.
- (6) Enzyme-substrate complex transition state stabilised products released; enzyme unchanged.

How to Outstand:

- EXAMINER TRAP: Never say enzymes 'destroy' or 'remove' activation energy — they LOWER it via an alternative reaction pathway.
- Cambridge awards marks for complementary shape AND CHARGE — not just shape.
- For induced-fit: state conformational change OPTIMALLY POSITIONS CATALYTIC RESIDUES — not just 'it changes shape'.
- Distinguish ESC (enzyme-substrate complex) from transition state — these are different.
- 'Describe' lock-and-key = rigid. 'Explain' induced-fit = WHY the change increases catalysis.

Resources: Q1–Q4 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 9 — Saturday, 03 Oct 2026

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 10 — Sunday, 04 Oct 2026

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 11 — Monday, 05 Oct 2026

Topic 3 — ENZYMES (9700 Syllabus 3.2a)

Today's Sub-topic: 3.2a Substrate concentration & K_m

Learning Objective: Apply Michaelis-Menten kinetics and define K_m as an inverse measure of enzyme-substrate affinity.

What to Learn:

- (1) Low $[S]$: rate proportional to $[S]$ = first-order kinetics.
- (2) High $[S]$: rate plateaus at V_{max} = zero-order kinetics; all active sites occupied simultaneously.
- (3) $K_m = [S]$ at which rate = $\frac{1}{2}V_{max}$. Low K_m = high affinity. High K_m = low affinity.
- (4) Calculate initial rate by drawing tangent at $t=0$ on progress curve; gradient = rate.
- (5) Read K_m from graph: draw horizontal at $\frac{1}{2}V_{max}$ read x-axis.
- (6) V_{max} is an asymptote — never actually reached at finite $[S]$.

How to Outstand:

- EXAMINER TRAP: K_m is NOT the $[S]$ at saturation — it is specifically the $[S]$ at HALF V_{max} .
- Always include UNITS of K_m : mol dm^{-3} .
- V_{max} plateau = all active sites occupied — NOT 'no more substrate'.
- Calculate initial rate using tangent method — state you draw the tangent.
- A* link: K_m approximates K_d of ESC when product formation is rate-limiting.

Resources: Q5–Q9 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 12 — Tuesday, 06 Oct 2026

Topic 3 — ENZYMES (9700 Syllabus 3.2b)

Today's Sub-topic: 3.2b Temperature and pH

Learning Objective: Explain at molecular level how temperature and pH alter enzyme activity through bond disruption.

What to Learn:

- (1) Temperature below optimum: increasing KE more frequent collisions rate rises. $Q_{10} \approx 2$.
- (2) Temperature above optimum: excess thermal energy disrupts hydrogen bonds and ionic bonds maintaining tertiary structure active site shape changes substrate cannot fit denaturation (IRREVERSIBLE).
- (3) pH effect: H/OH alter ionisation state of R-groups (amino acid side chains) lining active site charge and shape change substrate no longer complementary.
- (4) Extreme pH: irreversible denaturation.
- (5) Different enzymes have different pH optima: pepsin pH 2, salivary amylase pH 7, trypsin pH 8.

How to Outstand:

- EXAMINER TRAP: Do NOT say enzyme is 'killed' by heat — say HIGH TEMPERATURE DENATURES the enzyme by BREAKING HYDROGEN BONDS and IONIC BONDS maintaining the tertiary structure.
- Link temperature specifically to BOND TYPES: H-bonds and ionic bonds disrupted — NOT covalent bonds.
- pH: IONISATION STATE OF R-GROUPS changes — state this phrase.
- Denaturation = irreversible; inhibition = reversible — distinguish these.
- $Q_{10} = \frac{\text{rate}(T+10^{\circ}\text{C})}{\text{rate}(T^{\circ}\text{C})} \approx 2$ — know formula for calculation questions.

Resources: Q10–Q14 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 13 — Wednesday, 07 Oct 2026

Topic 3 — ENZYMES (9700 Syllabus 3.2c)

Today's Sub-topic: 3.2c Inhibitors

Learning Objective: Distinguish competitive and non-competitive inhibition kinetically using K_m and V_{max} data.

What to Learn:

- (1) Competitive inhibitor: structurally similar to substrate; binds ACTIVE SITE reversibly; blocks substrate; overcome by excess $[S]$. K_m INCREASES; V_{max} UNCHANGED.
- (2) Non-competitive: binds ALLOSTERIC SITE; conformational change distorts active site; cannot be overcome. V_{max} DECREASES; K_m UNCHANGED.
- (3) Examples: malonate vs succinate dehydrogenase (competitive); heavy metal Hg^2 on SH groups (irreversible non-competitive).
- (4) Allosteric end-product feedback inhibition (e.g. ATP inhibiting phosphofructokinase).
- (5) Lineweaver-Burk plot: competitive shifts x-intercept ($-1/K_m$ changes); non-competitive shifts y-intercept ($1/V_{max}$ changes).

How to Outstand:

- EXAMINER TRAP: Mnemonic for graphs: Competitive = K_m , V_{max} same. Non-competitive = V_{max} , K_m same. Candidates frequently swap these.
- Competitive inhibitor is STRUCTURALLY SIMILAR (complementary) to substrate — not identical.
- Allosteric site is SEPARATE and DISTINCT from active site — state this.
- Heavy metal ions form COVALENT bonds with SH groups IRREVERSIBLE (cannot be diluted away).
- Lineweaver-Burk: know x-intercept = $-1/K_m$, y-intercept = $1/V_{max}$.

Resources: Q15–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 14 — Thursday, 08 Oct 2026

Topic 3 — ENZYMES (9700 Syllabus 3.3)

Today's Sub-topic: 3.3 Enzyme technology

Learning Objective: Explain immobilisation methods, advantages, and industrial/biosensor applications of immobilised enzymes.

What to Learn:

- (1) Immobilisation methods: entrapment in calcium alginate gel beads; covalent bonding to inert support; adsorption to surface; cross-linking with glutaraldehyde.
- (2) Advantages: REUSABLE (enzyme recovered after reaction); PURE PRODUCT (enzyme-free product); THERMOSTABLE (fixed conformation); continuous-flow processing; reduced cost per unit.
- (3) Industrial examples: lactase in alginate beads lactose-free milk; glucose isomerase fructose syrup production.
- (4) Biosensor: enzyme immobilised on electrode; reaction generates electrical signal [substrate]; e.g. glucose oxidase for diabetic blood glucose monitoring.
- (5) Alginate bead protocol: mix enzyme with sodium alginate drip into CaCl₂ gel beads form.

How to Outstand:

- EXAMINER TRAP: Immobilised enzymes are NOT faster — they are REUSABLE and give PURE PRODUCT. Never claim increased speed as an advantage.
- Cambridge mark schemes: state exactly — reusable, pure enzyme-free product, more thermostable, continuous-flow possible.
- Specify the method of immobilisation: entrapment (most common in Cambridge questions) vs covalent bonding.
- Biosensor chain: substrate → enzyme → product → electrical signal → readout. Know the sequence.
- Alginate bead formation: sodium alginate + enzyme dripped into calcium chloride Ca²⁺ cross-links alginate polymer beads trap enzyme.

Resources: Q21–Q25 Section B, Q41–Q45 Section C

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 3: 09 Oct 2026 – 15 Oct 2026

Topic: ENZYMES

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

Weekly Practice: Hamna_Bio_Topic3_ENZYMES.pdf (50 Qs, 220 Marks, 3 hrs)

DAY 15 — Friday, 09 Oct 2026

Topic 3 — ENZYMES (9700 Syllabus 3.1)

Today's Sub-topic: 3.1 Enzyme action

Learning Objective: Explain how enzymes lower activation energy and distinguish lock-and-key from induced-fit models.

What to Learn:

- (1) Activation energy: energy barrier reactants must overcome; enzymes provide alternative pathway of lower E_a .
- (2) Enzymes are globular proteins; active site is specific in shape AND charge — complementary to substrate.
- (3) Lock-and-key model: rigid active site, substrate fits precisely without conformational change.
- (4) Induced-fit model: flexible active site; substrate binding causes conformational change that OPTIMALLY POSITIONS CATALYTIC RESIDUES.
- (5) Evidence for induced-fit: X-ray crystallography of hexokinase shows closure around glucose.
- (6) Enzyme-substrate complex transition state stabilised products released; enzyme unchanged.

How to Outstand:

- EXAMINER TRAP: Never say enzymes 'destroy' or 'remove' activation energy — they LOWER it via an alternative reaction pathway.
- Cambridge awards marks for complementary shape AND CHARGE — not just shape.
- For induced-fit: state conformational change OPTIMALLY POSITIONS CATALYTIC RESIDUES — not just 'it changes shape'.
- Distinguish ESC (enzyme-substrate complex) from transition state — these are different.
- 'Describe' lock-and-key = rigid. 'Explain' induced-fit = WHY the change increases catalysis.

Resources: Q1–Q4 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 16 — Saturday, 10 Oct 2026

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 17 — Sunday, 11 Oct 2026

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 18 — Monday, 12 Oct 2026

Topic 4 — CELL MEMBRANES AND TRANSPORT (9700 Syllabus 4.1)

Today's Sub-topic: 4.1 Fluid mosaic model

Learning Objective: Describe fluid mosaic membrane structure and explain cholesterol's bidirectional fluidity function.

What to Learn:

- (1) Phospholipid bilayer: 2 fatty acid tails (hydrophobic) + phosphate head (hydrophilic); spontaneous self-assembly due to hydrophobic effect.
- (2) Fluid: lipids and proteins can diffuse laterally within bilayer leaflet; confirmed by FRAP experiment.
- (3) Mosaic: integral proteins (span bilayer via transmembrane α -helices), peripheral proteins (surface-attached).
- (4) Glycoproteins (protein + carbohydrate) and glycolipids on extracellular surface ONLY — cell recognition, receptors, antigens.
- (5) Cholesterol: intercalated between tails; at HIGH temp reduces fluidity (prevents excessive lateral flow); at LOW temp prevents solidification. Membrane is ASYMMETRIC.

How to Outstand:

- EXAMINER TRAP: Cholesterol acts in BOTH directions — reduces fluidity at HIGH temperature AND prevents solidification at LOW temperature. Cambridge tests both.
- FLUID = lateral movement; MOSAIC = varied protein distribution — define each word separately.
- Carbohydrates are ONLY on EXTRACELLULAR face — never cytoplasmic. State the asymmetry.
- Integral proteins have hydrophobic transmembrane α -helices — give structural detail.
- FRAP: laser bleaches fluorescent lipids rate of fluorescence recovery proves lateral diffusion.

Resources: Q1–Q4 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 19 — Tuesday, 13 Oct 2026

Topic 4 — CELL MEMBRANES AND TRANSPORT (9700 Syllabus 4.2a)

Today's Sub-topic: 4.2a Diffusion and facilitated diffusion

Learning Objective: Distinguish simple and facilitated diffusion and apply Fick's law factors to biological exchange surfaces.

What to Learn:

- (1) Simple diffusion: small lipid-soluble molecules (O, CO, steroid hormones); down gradient; no proteins; no ATP.
- (2) Fick's Law: rate $(SA \times C) / \text{diffusion distance}$ — all three optimised in lungs (large SA, ventilation+perfusion maintain C, 0.5 μm barrier).
- (3) Facilitated via channel proteins: hydrophilic pores, ion-specific, gated (voltage-gated or ligand-gated); e.g. aquaporins (water), K channels.
- (4) Facilitated via carrier proteins: bind specific molecule; conformational change moves molecule across; e.g. GLUT2 (glucose).
- (5) Both types: down gradient, no ATP — only difference is protein carriers required.

How to Outstand:

- EXAMINER TRAP: 'Facilitated diffusion needs carrier proteins' is incomplete — specify CHANNEL proteins (ions, water) vs CARRIER proteins (glucose, amino acids). Cambridge marks them separately.
- 'From HIGH to LOW concentration' — not 'dilute to concentrated'.
- Aquaporins: water channel proteins — know the name.
- Channel proteins are GATED — they open and close; carrier proteins undergo conformational change.
- Apply Fick's Law to real examples: name all three factors and how each is maximised.

Resources: Q5–Q9 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 20 — Wednesday, 14 Oct 2026

Topic 4 — CELL MEMBRANES AND TRANSPORT (9700 Syllabus 4.2b)

Today's Sub-topic: 4.2b Osmosis and water potential

Learning Objective: Apply the water potential equation to predict osmotic movement in plant and animal cells.

What to Learn:

- (1) Osmosis: net movement of water from HIGH to LOW across selectively permeable membrane.
- (2) $\psi = s + p$. s always negative (solute potential). p positive in turgid cell, zero at incipient plasmolysis. Pure water $\psi = 0$.
- (3) Plant in hypotonic solution: water enters p rises fully turgid when $\psi = 0$ and net water movement stops.
- (4) Plasmolysis: hypertonic solution water leaves $p = 0$ membrane pulls from wall incipient plasmolysis.
- (5) Animal cells: no cell wall in hypotonic: HAEMOLYSIS (RBCs burst). In hypertonic: CRENATION (RBC shrinks). Units: kPa.

How to Outstand:

- EXAMINER TRAP: NEVER say 'water moves from dilute to concentrated'. Say: 'water moves from HIGH water potential to LOW water potential'. Cambridge deducts marks for this.
- At full turgor: p is maximum, $\psi = 0$. At incipient plasmolysis: $p = 0$, $\psi = s$. Know both conditions.
- Always include units: kPa.
- Compare ψ of cell vs external solution to determine direction of water movement.
- Haemolysis (hypotonic) vs Crenation (hypertonic) — use correct vocabulary for red blood cells.

Resources: Q10–Q15 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 21 — Thursday, 15 Oct 2026

Topic 4 — CELL MEMBRANES AND TRANSPORT (9700 Syllabus 4.2c)

Today's Sub-topic: 4.2c Active transport and bulk transport

Learning Objective: Explain primary and secondary active transport and describe endocytosis/exocytosis mechanisms.

What to Learn:

- (1) Primary active transport: uses ATP directly; e.g. Na/K-ATPase: 3 Na OUT, 2 K IN per ATP hydrolysed; maintains resting membrane potential.
- (2) Secondary active transport (cotransport): uses electrochemical gradient from primary transport; no direct ATP; e.g. Na/glucose symporter in ileum — Na moves down gradient, drags glucose AGAINST gradient into cell.
- (3) Key feature: against concentration gradient + requires energy.
- (4) Phagocytosis: pseudopodia extend around solid particle membrane fuses phagosome forms.
- (5) Pinocytosis: fluid engulfed in small vesicles.
- (6) Exocytosis: secretory vesicles from Golgi fuse with membrane contents released; requires ATP + microtubules.

How to Outstand:

- EXAMINER TRAP: 'Requires energy' is vague. Specify: ATP directly hydrolysed (primary) OR pre-existing electrochemical gradient used (secondary).
- Na/K-ATPase stoichiometry: 3 Na out, 2 K in per ATP — memorise.
- Secondary transport: the Na gradient is the DRIVING FORCE — it was established by ATP-dependent primary transport.
- Exocytosis is NOT passive: requires microtubules, ATP, and vesicle-membrane fusion proteins (SNAREs).
- Symport = same direction; antiport = opposite direction. Know both terms.

Resources: Q16–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 4: 16 Oct 2026 – 22 Oct 2026

Topic: CELL MEMBRANES AND TRANSPORT

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

DAY 22 — Friday, 16 Oct 2026

Topic 4 — CELL MEMBRANES AND TRANSPORT (9700 Syllabus 4.3)

Today's Sub-topic: 4.3 Practical investigations

Learning Objective: Apply membrane permeability, osmosis, and SA:V diffusion practical protocols with correct analysis.

What to Learn:

- (1) Beetroot permeability: equal-sized cylinders, wash in running water first, heat in water baths at varied temperatures, measure betacyanin absorbance at 600nm by colorimeter, higher absorbance = more permeable.
- (2) Osmosis in potato chips: immerse equal-mass chips in graded sucrose solutions, measure % mass change after 30 min, plot vs [sucrose], x-intercept = isotonic point = tissue .
- (3) SA:V agar blocks: different cube sizes immersed in acid; time for acid to reach centre; larger blocks = longer time AND smaller % volume stained; demonstrates SA:V limitation.
- (4) Controls: cylinder size, time, temperature, colorimeter blanking.
- (5) Statistics: mean, standard deviation, t-test.

How to Outstand:

- EXAMINER TRAP: WASH beetroot cylinders in running water first — this removes pigment from cut surface-damaged cells. Missing this step loses marks.
- Always calculate % CHANGE in mass (not absolute), because initial masses differ.
- The x-intercept of % mass change graph = isotonic point = water potential of tissue (). Examiners expect this interpretation.
- Colorimeter must be BLANKED (zeroed) with distilled water — always state this.
- SA:V agar conclusion: larger organisms cannot rely solely on diffusion — must have specialised transport systems. State this conclusion.

Resources: Q21–Q25 Section B

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 23 — Saturday, 17 Oct 2026

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 24 — Sunday, 18 Oct 2026

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 25 — Monday, 19 Oct 2026

Topic 4 — CELL MEMBRANES AND TRANSPORT (9700 Syllabus 4.1)

Today's Sub-topic: 4.1 Fluid mosaic model

Learning Objective: Describe fluid mosaic membrane structure and explain cholesterol's bidirectional fluidity function.

What to Learn:

- (1) Phospholipid bilayer: 2 fatty acid tails (hydrophobic) + phosphate head (hydrophilic); spontaneous self-assembly due to hydrophobic effect.
- (2) Fluid: lipids and proteins can diffuse laterally within bilayer leaflet; confirmed by FRAP experiment.
- (3) Mosaic: integral proteins (span bilayer via transmembrane α -helices), peripheral proteins (surface-attached).
- (4) Glycoproteins (protein + carbohydrate) and glycolipids on extracellular surface ONLY — cell recognition, receptors, antigens.
- (5) Cholesterol: intercalated between tails; at HIGH temp reduces fluidity (prevents excessive lateral flow); at LOW temp prevents solidification. Membrane is ASYMMETRIC.

How to Outstand:

- EXAMINER TRAP: Cholesterol acts in BOTH directions — reduces fluidity at HIGH temperature AND prevents solidification at LOW temperature. Cambridge tests both.
- FLUID = lateral movement; MOSAIC = varied protein distribution — define each word separately.
- Carbohydrates are ONLY on EXTRACELLULAR face — never cytoplasmic. State the asymmetry.
- Integral proteins have hydrophobic transmembrane α -helices — give structural detail.
- FRAP: laser bleaches fluorescent lipids rate of fluorescence recovery proves lateral diffusion.

Resources: Q1–Q4 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 26 — Tuesday, 20 Oct 2026

Topic 4 — CELL MEMBRANES AND TRANSPORT (9700 Syllabus 4.2a)

Today's Sub-topic: 4.2a Diffusion and facilitated diffusion

Learning Objective: Distinguish simple and facilitated diffusion and apply Fick's law factors to biological exchange surfaces.

What to Learn:

- (1) Simple diffusion: small lipid-soluble molecules (O, CO, steroid hormones); down gradient; no proteins; no ATP.
- (2) Fick's Law: rate $(SA \times C) / \text{diffusion distance}$ — all three optimised in lungs (large SA, ventilation+perfusion maintain C, 0.5 μm barrier).
- (3) Facilitated via channel proteins: hydrophilic pores, ion-specific, gated (voltage-gated or ligand-gated); e.g. aquaporins (water), K channels.
- (4) Facilitated via carrier proteins: bind specific molecule; conformational change moves molecule across; e.g. GLUT2 (glucose).
- (5) Both types: down gradient, no ATP — only difference is protein carriers required.

How to Outstand:

- EXAMINER TRAP: 'Facilitated diffusion needs carrier proteins' is incomplete — specify CHANNEL proteins (ions, water) vs CARRIER proteins (glucose, amino acids). Cambridge marks them separately.
- 'From HIGH to LOW concentration' — not 'dilute to concentrated'.
- Aquaporins: water channel proteins — know the name.
- Channel proteins are GATED — they open and close; carrier proteins undergo conformational change.
- Apply Fick's Law to real examples: name all three factors and how each is maximised.

Resources: Q5–Q9 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 27 — Wednesday, 21 Oct 2026

Topic 4 — CELL MEMBRANES AND TRANSPORT (9700 Syllabus 4.2b)

Today's Sub-topic: 4.2b Osmosis and water potential

Learning Objective: Apply the water potential equation to predict osmotic movement in plant and animal cells.

What to Learn:

- (1) Osmosis: net movement of water from HIGH to LOW across selectively permeable membrane.
- (2) $\psi = s + p$. s always negative (solute potential). p positive in turgid cell, zero at incipient plasmolysis. Pure water $\psi = 0$.
- (3) Plant in hypotonic solution: water enters p rises fully turgid when $\psi = 0$ and net water movement stops.
- (4) Plasmolysis: hypertonic solution water leaves $p = 0$ membrane pulls from wall incipient plasmolysis.
- (5) Animal cells: no cell wall in hypotonic: HAEMOLYSIS (RBCs burst). In hypertonic: CRENATION (RBC shrinks). Units: kPa.

How to Outstand:

- EXAMINER TRAP: NEVER say 'water moves from dilute to concentrated'. Say: 'water moves from HIGH water potential to LOW water potential'. Cambridge deducts marks for this.
- At full turgor: p is maximum, $\psi = 0$. At incipient plasmolysis: $p = 0$, $\psi = s$. Know both conditions.
- Always include units: kPa.
- Compare ψ of cell vs external solution to determine direction of water movement.
- Haemolysis (hypotonic) vs Crenation (hypertonic) — use correct vocabulary for red blood cells.

Resources: Q10–Q15 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 28 — Thursday, 22 Oct 2026

Topic 4 — CELL MEMBRANES AND TRANSPORT (9700 Syllabus 4.2c)

Today's Sub-topic: 4.2c Active transport and bulk transport

Learning Objective: Explain primary and secondary active transport and describe endocytosis/exocytosis mechanisms.

What to Learn:

- (1) Primary active transport: uses ATP directly; e.g. Na/K-ATPase: 3 Na OUT, 2 K IN per ATP hydrolysed; maintains resting membrane potential.
- (2) Secondary active transport (cotransport): uses electrochemical gradient from primary transport; no direct ATP; e.g. Na/glucose symporter in ileum — Na moves down gradient, drags glucose AGAINST gradient into cell.
- (3) Key feature: against concentration gradient + requires energy.
- (4) Phagocytosis: pseudopodia extend around solid particle membrane fuses phagosome forms.
- (5) Pinocytosis: fluid engulfed in small vesicles.
- (6) Exocytosis: secretory vesicles from Golgi fuse with membrane contents released; requires ATP + microtubules.

How to Outstand:

- EXAMINER TRAP: 'Requires energy' is vague. Specify: ATP directly hydrolysed (primary) OR pre-existing electrochemical gradient used (secondary).
- Na/K-ATPase stoichiometry: 3 Na out, 2 K in per ATP — memorise.
- Secondary transport: the Na gradient is the DRIVING FORCE — it was established by ATP-dependent primary transport.
- Exocytosis is NOT passive: requires microtubules, ATP, and vesicle-membrane fusion proteins (SNAREs).
- Symport = same direction; antiport = opposite direction. Know both terms.

Resources: Q16–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 5: 23 Oct 2026 – 29 Oct 2026

Topic: CELL MEMBRANES AND TRANSPORT

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

DAY 29 — Friday, 23 Oct 2026

Topic 4 — CELL MEMBRANES AND TRANSPORT (9700 Syllabus 4.3)

Today's Sub-topic: 4.3 Practical investigations

Learning Objective: Apply membrane permeability, osmosis, and SA:V diffusion practical protocols with correct analysis.

What to Learn:

- (1) Beetroot permeability: equal-sized cylinders, wash in running water first, heat in water baths at varied temperatures, measure betacyanin absorbance at 600nm by colorimeter, higher absorbance = more permeable.
- (2) Osmosis in potato chips: immerse equal-mass chips in graded sucrose solutions, measure % mass change after 30 min, plot vs [sucrose], x-intercept = isotonic point = tissue .
- (3) SA:V agar blocks: different cube sizes immersed in acid; time for acid to reach centre; larger blocks = longer time AND smaller % volume stained; demonstrates SA:V limitation.
- (4) Controls: cylinder size, time, temperature, colorimeter blanking.
- (5) Statistics: mean, standard deviation, t-test.

How to Outstand:

- EXAMINER TRAP: WASH beetroot cylinders in running water first — this removes pigment from cut surface-damaged cells. Missing this step loses marks.
- Always calculate % CHANGE in mass (not absolute), because initial masses differ.
- The x-intercept of % mass change graph = isotonic point = water potential of tissue (). Examiners expect this interpretation.
- Colorimeter must be BLANKED (zeroed) with distilled water — always state this.
- SA:V agar conclusion: larger organisms cannot rely solely on diffusion — must have specialised transport systems. State this conclusion.

Resources: Q21–Q25 Section B

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 30 — Saturday, 24 Oct 2026

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 31 — Sunday, 25 Oct 2026

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 32 — Monday, 26 Oct 2026

Topic 4 — CELL MEMBRANES AND TRANSPORT (9700 Syllabus 4.1)

Today's Sub-topic: 4.1 Fluid mosaic model

Learning Objective: Describe fluid mosaic membrane structure and explain cholesterol's bidirectional fluidity function.

What to Learn:

- (1) Phospholipid bilayer: 2 fatty acid tails (hydrophobic) + phosphate head (hydrophilic); spontaneous self-assembly due to hydrophobic effect.
- (2) Fluid: lipids and proteins can diffuse laterally within bilayer leaflet; confirmed by FRAP experiment.
- (3) Mosaic: integral proteins (span bilayer via transmembrane α -helices), peripheral proteins (surface-attached).
- (4) Glycoproteins (protein + carbohydrate) and glycolipids on extracellular surface ONLY — cell recognition, receptors, antigens.
- (5) Cholesterol: intercalated between tails; at HIGH temp reduces fluidity (prevents excessive lateral flow); at LOW temp prevents solidification. Membrane is ASYMMETRIC.

How to Outstand:

- EXAMINER TRAP: Cholesterol acts in BOTH directions — reduces fluidity at HIGH temperature AND prevents solidification at LOW temperature. Cambridge tests both.
- FLUID = lateral movement; MOSAIC = varied protein distribution — define each word separately.
- Carbohydrates are ONLY on EXTRACELLULAR face — never cytoplasmic. State the asymmetry.
- Integral proteins have hydrophobic transmembrane α -helices — give structural detail.
- FRAP: laser bleaches fluorescent lipids rate of fluorescence recovery proves lateral diffusion.

Resources: Q1–Q4 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 33 — Tuesday, 27 Oct 2026

Topic 4 — CELL MEMBRANES AND TRANSPORT (9700 Syllabus 4.2a)

Today's Sub-topic: 4.2a Diffusion and facilitated diffusion

Learning Objective: Distinguish simple and facilitated diffusion and apply Fick's law factors to biological exchange surfaces.

What to Learn:

- (1) Simple diffusion: small lipid-soluble molecules (O, CO, steroid hormones); down gradient; no proteins; no ATP.
- (2) Fick's Law: $\text{rate} = \frac{\text{SA} \times C}{\text{diffusion distance}}$ — all three optimised in lungs (large SA, ventilation+perfusion maintain C, 0.5 μm barrier).
- (3) Facilitated via channel proteins: hydrophilic pores, ion-specific, gated (voltage-gated or ligand-gated); e.g. aquaporins (water), K channels.
- (4) Facilitated via carrier proteins: bind specific molecule; conformational change moves molecule across; e.g. GLUT2 (glucose).
- (5) Both types: down gradient, no ATP — only difference is protein carriers required.

How to Outstand:

- EXAMINER TRAP: 'Facilitated diffusion needs carrier proteins' is incomplete — specify CHANNEL proteins (ions, water) vs CARRIER proteins (glucose, amino acids). Cambridge marks them separately.
- 'From HIGH to LOW concentration' — not 'dilute to concentrated'.
- Aquaporins: water channel proteins — know the name.
- Channel proteins are GATED — they open and close; carrier proteins undergo conformational change.
- Apply Fick's Law to real examples: name all three factors and how each is maximised.

Resources: Q5–Q9 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 34 — Wednesday, 28 Oct 2026

Topic 4 — CELL MEMBRANES AND TRANSPORT (9700 Syllabus 4.2b)

Today's Sub-topic: 4.2b Osmosis and water potential

Learning Objective: Apply the water potential equation to predict osmotic movement in plant and animal cells.

What to Learn:

- (1) Osmosis: net movement of water from HIGH to LOW across selectively permeable membrane.
- (2) $\psi = s + p$. s always negative (solute potential). p positive in turgid cell, zero at incipient plasmolysis. Pure water $\psi = 0$.
- (3) Plant in hypotonic solution: water enters p rises fully turgid when $\psi = 0$ and net water movement stops.
- (4) Plasmolysis: hypertonic solution water leaves $p = 0$ membrane pulls from wall incipient plasmolysis.
- (5) Animal cells: no cell wall in hypotonic: HAEMOLYSIS (RBCs burst). In hypertonic: CRENATION (RBC shrinks). Units: kPa.

How to Outstand:

- EXAMINER TRAP: NEVER say 'water moves from dilute to concentrated'. Say: 'water moves from HIGH water potential to LOW water potential'. Cambridge deducts marks for this.
- At full turgor: p is maximum, $\psi = 0$. At incipient plasmolysis: $p = 0$, $\psi = s$. Know both conditions.
- Always include units: kPa.
- Compare ψ of cell vs external solution to determine direction of water movement.
- Haemolysis (hypotonic) vs Crenation (hypertonic) — use correct vocabulary for red blood cells.

Resources: Q10–Q15 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 35 — Thursday, 29 Oct 2026

Topic 4 — CELL MEMBRANES AND TRANSPORT (9700 Syllabus 4.2c)

Today's Sub-topic: 4.2c Active transport and bulk transport

Learning Objective: Explain primary and secondary active transport and describe endocytosis/exocytosis mechanisms.

What to Learn:

- (1) Primary active transport: uses ATP directly; e.g. Na/K-ATPase: 3 Na OUT, 2 K IN per ATP hydrolysed; maintains resting membrane potential.
- (2) Secondary active transport (cotransport): uses electrochemical gradient from primary transport; no direct ATP; e.g. Na/glucose symporter in ileum — Na moves down gradient, drags glucose AGAINST gradient into cell.
- (3) Key feature: against concentration gradient + requires energy.
- (4) Phagocytosis: pseudopodia extend around solid particle membrane fuses phagosome forms.
- (5) Pinocytosis: fluid engulfed in small vesicles.
- (6) Exocytosis: secretory vesicles from Golgi fuse with membrane contents released; requires ATP + microtubules.

How to Outstand:

- EXAMINER TRAP: 'Requires energy' is vague. Specify: ATP directly hydrolysed (primary) OR pre-existing electrochemical gradient used (secondary).
- Na/K-ATPase stoichiometry: 3 Na out, 2 K in per ATP — memorise.
- Secondary transport: the Na gradient is the DRIVING FORCE — it was established by ATP-dependent primary transport.
- Exocytosis is NOT passive: requires microtubules, ATP, and vesicle-membrane fusion proteins (SNAREs).
- Symport = same direction; antiport = opposite direction. Know both terms.

Resources: Q16–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 6: 30 Oct 2026 – 05 Nov 2026

Topic: CELL MEMBRANES AND TRANSPORT

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

Weekly Practice: Hamna_Bio_Topic4_CELL_MEMBRANES_AND_TRANSPORT.pdf (50 Qs, 220 Marks, 3 hrs)

DAY 36 — Friday, 30 Oct 2026

Topic 4 — CELL MEMBRANES AND TRANSPORT (9700 Syllabus 4.3)

Today's Sub-topic: 4.3 Practical investigations

Learning Objective: Apply membrane permeability, osmosis, and SA:V diffusion practical protocols with correct analysis.

What to Learn:

- (1) Beetroot permeability: equal-sized cylinders, wash in running water first, heat in water baths at varied temperatures, measure betacyanin absorbance at 600nm by colorimeter, higher absorbance = more permeable.
- (2) Osmosis in potato chips: immerse equal-mass chips in graded sucrose solutions, measure % mass change after 30 min, plot vs [sucrose], x-intercept = isotonic point = tissue .
- (3) SA:V agar blocks: different cube sizes immersed in acid; time for acid to reach centre; larger blocks = longer time AND smaller % volume stained; demonstrates SA:V limitation.
- (4) Controls: cylinder size, time, temperature, colorimeter blanking.
- (5) Statistics: mean, standard deviation, t-test.

How to Outstand:

- EXAMINER TRAP: WASH beetroot cylinders in running water first — this removes pigment from cut surface-damaged cells. Missing this step loses marks.
- Always calculate % CHANGE in mass (not absolute), because initial masses differ.
- The x-intercept of % mass change graph = isotonic point = water potential of tissue (). Examiners expect this interpretation.
- Colorimeter must be BLANKED (zeroed) with distilled water — always state this.
- SA:V agar conclusion: larger organisms cannot rely solely on diffusion — must have specialised transport systems. State this conclusion.

Resources: Q21–Q25 Section B

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 37 — Saturday, 31 Oct 2026

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 38 — Sunday, 01 Nov 2026

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 39 — Monday, 02 Nov 2026

Topic 5 — THE MITOTIC CELL CYCLE (9700 Syllabus 5.1)

Today's Sub-topic: 5.1 Interphase and DNA replication

Learning Objective: Describe interphase events and explain semi-conservative replication via Meselson–Stahl.

What to Learn:

- (1) G1: cell growth, protein synthesis, organelle biogenesis. S: DNA replication, 2C → 4C. G2: organelle replication, tubulin synthesis.
- (2) Semi-conservative: each new helix has one parental + one new strand.
- (3) Meselson–Stahl: bacteria in ¹⁵N transfer to ¹⁴N after 1 replication: ALL intermediate density; after 2: half intermediate + half light proves semi-conservative.
- (4) Enzymes: helicase (unwinds), primase (RNA primer), DNA pol III (5'3' elongation), pol I (removes primer), ligase (seals nicks).
- (5) Leading strand: continuous; lagging strand: Okazaki fragments.
- (6) After replication: 2n maintained; each chromosome = 2 sister chromatids.

How to Outstand:

- EXAMINER TRAP: DNA replication is in S PHASE of INTERPHASE — NOT in mitosis. This is Cambridge's most penalised factual error on this topic.
- Meselson–Stahl: know what CsCl density gradient ultracentrifugation measures and what each band pattern proves.
- DNA polymerase only works 5'3' — this CAUSES lagging strand discontinuity.
- Distinguish semi-conservative from conservative.
- C-value: 2C in G1, 4C after S phase.

Resources: Q1–Q4 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 40 — Tuesday, 03 Nov 2026

Topic 5 — THE MITOTIC CELL CYCLE (9700 Syllabus 5.2)

Today's Sub-topic: 5.2 Stages of mitosis

Learning Objective: Describe precise chromosomal events in each mitotic stage and apply experimental techniques.

What to Learn:

- (1) Prophase: chromatin condenses, nuclear envelope breaks down, spindle forms, kinetochores form.
- (2) Metaphase: chromosomes align at metaphase plate; spindle fibres attach.
- (3) Anaphase: centromeres split; chromatids pulled to poles.
- (4) Telophase: nuclear envelopes reform, chromosomes decondense.
- (5) Cytokinesis: animal = cleavage furrow; plant = cell plate.
- (6) Allium root tip squash technique; Mitotic Index = $(\text{cells in mitosis} / \text{total}) \times 100\%$.

How to Outstand:

- EXAMINER TRAP: In ANAPHASE the CHROMOSOME NUMBER per HALF-CELL doubles.
- Spindle fibres attach to KINETOCHORE.
- Colchicine arrests at METAPHASE.
- High MI = rapidly dividing tissue.

Resources: Q5–Q10 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 41 — Wednesday, 04 Nov 2026

Topic 5 — THE MITOTIC CELL CYCLE (9700 Syllabus 5.3)

Today's Sub-topic: 5.3 Cell cycle control and cancer

Learning Objective: Explain checkpoint mechanisms and multi-step carcinogenesis via oncogenes and tumour suppressor mutations.

What to Learn:

- (1) Checkpoints: G1/S, G2/M, SAC.
- (2) Proto-oncogenes oncogenes: gain-of-function mutation; dominant.
- (3) Tumour suppressor genes (TP53, Rb): loss of function; recessive.
- (4) Multi-step carcinogenesis: 5–7 mutations required.
- (5) Cancer properties: uncontrolled mitosis, angiogenesis, metastasis.
- (6) Stem cells: totipotent, pluripotent, multipotent.

How to Outstand:

- EXAMINER TRAP: Cancer is NOT 'cells dividing faster' — it is UNCONTROLLED division.
- TP53 is 'guardian of the genome'.
- Oncogene = dominant vs tumour suppressor = recessive.
- VEGF stimulates angiogenesis.
- SAC failure aneuploidy.

Resources: Q11–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 42 — Thursday, 05 Nov 2026

Topic 5 — THE MITOTIC CELL CYCLE (9700 Syllabus 5.1)

Today's Sub-topic: 5.1 Interphase and DNA replication

Learning Objective: Describe interphase events and explain semi-conservative replication via Meselson–Stahl.

What to Learn:

- (1) G1: cell growth, protein synthesis, organelle biogenesis. S: DNA replication, 2C → 4C. G2: organelle replication, tubulin synthesis.
- (2) Semi-conservative: each new helix has one parental + one new strand.
- (3) Meselson–Stahl: bacteria in ¹⁵N transfer to ¹⁴N after 1 replication: ALL intermediate density; after 2: half intermediate + half light proves semi-conservative.
- (4) Enzymes: helicase (unwinds), primase (RNA primer), DNA pol III (5'3' elongation), pol I (removes primer), ligase (seals nicks).
- (5) Leading strand: continuous; lagging strand: Okazaki fragments.
- (6) After replication: 2n maintained; each chromosome = 2 sister chromatids.

How to Outstand:

- EXAMINER TRAP: DNA replication is in S PHASE of INTERPHASE — NOT in mitosis. This is Cambridge's most penalised factual error on this topic.
- Meselson–Stahl: know what CsCl density gradient ultracentrifugation measures and what each band pattern proves.
- DNA polymerase only works 5'3' — this CAUSES lagging strand discontinuity.
- Distinguish semi-conservative from conservative.
- C-value: 2C in G1, 4C after S phase.

Resources: Q1–Q4 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 7: 06 Nov 2026 – 12 Nov 2026

Topic: THE MITOTIC CELL CYCLE

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

DAY 43 — Friday, 06 Nov 2026

Topic 5 — THE MITOTIC CELL CYCLE (9700 Syllabus 5.2)

Today's Sub-topic: 5.2 Stages of mitosis

Learning Objective: Describe precise chromosomal events in each mitotic stage and apply experimental techniques.

What to Learn:

- (1) Prophase: chromatin condenses, nuclear envelope breaks down, spindle forms, kinetochores form.
- (2) Metaphase: chromosomes align at metaphase plate; spindle fibres attach.
- (3) Anaphase: centromeres split; chromatids pulled to poles.
- (4) Telophase: nuclear envelopes reform, chromosomes decondense.
- (5) Cytokinesis: animal = cleavage furrow; plant = cell plate.
- (6) Allium root tip squash technique; Mitotic Index = $(\text{cells in mitosis} / \text{total}) \times 100\%$.

How to Outstand:

- EXAMINER TRAP: In ANAPHASE the CHROMOSOME NUMBER per HALF-CELL doubles.
- Spindle fibres attach to KINETOCHORE.
- Colchicine arrests at METAPHASE.
- High MI = rapidly dividing tissue.

Resources: Q5–Q10 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 44 — Saturday, 07 Nov 2026

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 45 — Sunday, 08 Nov 2026

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 46 — Monday, 09 Nov 2026

Topic 5 — THE MITOTIC CELL CYCLE (9700 Syllabus 5.3)

Today's Sub-topic: 5.3 Cell cycle control and cancer

Learning Objective: Explain checkpoint mechanisms and multi-step carcinogenesis via oncogenes and tumour suppressor mutations.

What to Learn:

- (1) Checkpoints: G1/S, G2/M, SAC.
- (2) Proto-oncogenes oncogenes: gain-of-function mutation; dominant.
- (3) Tumour suppressor genes (TP53, Rb): loss of function; recessive.
- (4) Multi-step carcinogenesis: 5–7 mutations required.
- (5) Cancer properties: uncontrolled mitosis, angiogenesis, metastasis.
- (6) Stem cells: totipotent, pluripotent, multipotent.

How to Outstand:

- EXAMINER TRAP: Cancer is NOT 'cells dividing faster' — it is UNCONTROLLED division.
- TP53 is 'guardian of the genome'.
- Oncogene = dominant vs tumour suppressor = recessive.
- VEGF stimulates angiogenesis.
- SAC failure aneuploidy.

Resources: Q11–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 47 — Tuesday, 10 Nov 2026

Topic 5 — THE MITOTIC CELL CYCLE (9700 Syllabus 5.1)

Today's Sub-topic: 5.1 Interphase and DNA replication

Learning Objective: Describe interphase events and explain semi-conservative replication via Meselson–Stahl.

What to Learn:

- (1) G1: cell growth, protein synthesis, organelle biogenesis. S: DNA replication, 2C → 4C. G2: organelle replication, tubulin synthesis.
- (2) Semi-conservative: each new helix has one parental + one new strand.
- (3) Meselson–Stahl: bacteria in ¹⁵N transfer to ¹⁴N after 1 replication: ALL intermediate density; after 2: half intermediate + half light proves semi-conservative.
- (4) Enzymes: helicase (unwinds), primase (RNA primer), DNA pol III (5'3' elongation), pol I (removes primer), ligase (seals nicks).
- (5) Leading strand: continuous; lagging strand: Okazaki fragments.
- (6) After replication: 2n maintained; each chromosome = 2 sister chromatids.

How to Outstand:

- EXAMINER TRAP: DNA replication is in S PHASE of INTERPHASE — NOT in mitosis. This is Cambridge's most penalised factual error on this topic.
- Meselson–Stahl: know what CsCl density gradient ultracentrifugation measures and what each band pattern proves.
- DNA polymerase only works 5'3' — this CAUSES lagging strand discontinuity.
- Distinguish semi-conservative from conservative.
- C-value: 2C in G1, 4C after S phase.

Resources: Q1–Q4 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 48 — Wednesday, 11 Nov 2026

Topic 5 — THE MITOTIC CELL CYCLE (9700 Syllabus 5.2)

Today's Sub-topic: 5.2 Stages of mitosis

Learning Objective: Describe precise chromosomal events in each mitotic stage and apply experimental techniques.

What to Learn:

- (1) Prophase: chromatin condenses, nuclear envelope breaks down, spindle forms, kinetochores form.
- (2) Metaphase: chromosomes align at metaphase plate; spindle fibres attach.
- (3) Anaphase: centromeres split; chromatids pulled to poles.
- (4) Telophase: nuclear envelopes reform, chromosomes decondense.
- (5) Cytokinesis: animal = cleavage furrow; plant = cell plate.
- (6) Allium root tip squash technique; Mitotic Index = $(\text{cells in mitosis} / \text{total}) \times 100\%$.

How to Outstand:

- EXAMINER TRAP: In ANAPHASE the CHROMOSOME NUMBER per HALF-CELL doubles.
- Spindle fibres attach to KINETOCHORE.
- Colchicine arrests at METAPHASE.
- High MI = rapidly dividing tissue.

Resources: Q5–Q10 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 49 — Thursday, 12 Nov 2026

Topic 5 — THE MITOTIC CELL CYCLE (9700 Syllabus 5.3)

Today's Sub-topic: 5.3 Cell cycle control and cancer

Learning Objective: Explain checkpoint mechanisms and multi-step carcinogenesis via oncogenes and tumour suppressor mutations.

What to Learn:

- (1) Checkpoints: G1/S, G2/M, SAC.
- (2) Proto-oncogenes oncogenes: gain-of-function mutation; dominant.
- (3) Tumour suppressor genes (TP53, Rb): loss of function; recessive.
- (4) Multi-step carcinogenesis: 5–7 mutations required.
- (5) Cancer properties: uncontrolled mitosis, angiogenesis, metastasis.
- (6) Stem cells: totipotent, pluripotent, multipotent.

How to Outstand:

- EXAMINER TRAP: Cancer is NOT 'cells dividing faster' — it is UNCONTROLLED division.
- TP53 is 'guardian of the genome'.
- Oncogene = dominant vs tumour suppressor = recessive.
- VEGF stimulates angiogenesis.
- SAC failure aneuploidy.

Resources: Q11–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 8: 13 Nov 2026 – 19 Nov 2026

Topic: THE MITOTIC CELL CYCLE

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

Weekly Practice: Hamna_Bio_Topic5_THE_MITOTIC_CELL_CYCLE.pdf (50 Qs, 220 Marks, 3 hrs)

DAY 50 — Friday, 13 Nov 2026

Topic 5 — THE MITOTIC CELL CYCLE (9700 Syllabus 5.1)

Today's Sub-topic: 5.1 Interphase and DNA replication

Learning Objective: Describe interphase events and explain semi-conservative replication via Meselson–Stahl.

What to Learn:

- (1) G1: cell growth, protein synthesis, organelle biogenesis. S: DNA replication, 2C → 4C. G2: organelle replication, tubulin synthesis.
- (2) Semi-conservative: each new helix has one parental + one new strand.
- (3) Meselson–Stahl: bacteria in ¹⁵N transfer to ¹⁴N after 1 replication: ALL intermediate density; after 2: half intermediate + half light proves semi-conservative.
- (4) Enzymes: helicase (unwinds), primase (RNA primer), DNA pol III (5'3' elongation), pol I (removes primer), ligase (seals nicks).
- (5) Leading strand: continuous; lagging strand: Okazaki fragments.
- (6) After replication: 2n maintained; each chromosome = 2 sister chromatids.

How to Outstand:

- EXAMINER TRAP: DNA replication is in S PHASE of INTERPHASE — NOT in mitosis. This is Cambridge's most penalised factual error on this topic.
- Meselson–Stahl: know what CsCl density gradient ultracentrifugation measures and what each band pattern proves.
- DNA polymerase only works 5'3' — this CAUSES lagging strand discontinuity.
- Distinguish semi-conservative from conservative.
- C-value: 2C in G1, 4C after S phase.

Resources: Q1–Q4 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 51 — Saturday, 14 Nov 2026

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 52 — Sunday, 15 Nov 2026

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 53 — Monday, 16 Nov 2026

Topic 6 — NUCLEIC ACIDS AND PROTEIN SYNTHESIS (9700 Syllabus 6.1)

Today's Sub-topic: 6.1 DNA structure and replication

Learning Objective: Describe DNA double helix structure and semi-conservative replication with enzyme roles.

What to Learn:

- (1) Nucleotide: deoxyribose + phosphate + nitrogenous base.
- (2) Chargaff's rules: A=T, GC.
- (3) Double helix: antiparallel strands.
- (4) Replication enzymes: helicase, primase, DNA pol III, pol I, ligase.
- (5) Leading continuous, lagging Okazaki.
- (6) Nucleosome: DNA wound around histone octamer.

How to Outstand:

- Chargaff's calculation: if %G = 20% then %C = 20%, %A = %T = 30%.
- GC has 3 H-bonds.
- Antiparallel notation: draw 5'3' direction.
- Histones: positively charged proteins.

Resources: Q1–Q5 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 54 — Tuesday, 17 Nov 2026

Topic 6 — NUCLEIC ACIDS AND PROTEIN SYNTHESIS (9700 Syllabus 6.2a)

Today's Sub-topic: 6.2a Transcription

Learning Objective: Describe transcription mechanism including promoter, RNA polymerase, and eukaryotic mRNA processing.

What to Learn:

- (1) RNA pol binds promoter; synthesises mRNA 5'3'.
- (2) mRNA = same sequence as CODING strand.
- (3) Eukaryotic mRNA processing: 5' cap, poly-A tail, splicing.
- (4) RNA types: mRNA, tRNA, rRNA.
- (5) Prokaryotes: no processing; coupled transcription-translation.

How to Outstand:

- EXAMINER TRAP: mRNA has the SAME sequence as the CODING strand (with U for T).
- Template = antisense = 3'5'. Coding = sense = same as mRNA.
- Introns interrupt; exons are expressed.

Resources: Q6–Q10 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 55 — Wednesday, 18 Nov 2026

Topic 6 — NUCLEIC ACIDS AND PROTEIN SYNTHESIS (9700 Syllabus 6.2b)

Today's Sub-topic: 6.2b Translation

Learning Objective: Describe ribosome mechanism of translation including codon-anticodon pairing and peptide bond formation.

What to Learn:

- (1) Ribosome: 80S (eukaryotes), 70S (prokaryotes).
- (2) tRNA: anticodon loop, 3' CCA acceptor arm.
- (3) Translation: Initiation, Elongation, Termination.
- (4) Genetic code: triplet, non-overlapping, degenerate, universal, non-ambiguous.
- (5) Polysomes: multiple ribosomes on one mRNA.

How to Outstand:

- EXAMINER TRAP: Anticodon on tRNA is ANTIPARALLEL to mRNA codon.
- Genetic code is DEGENERATE but NON-AMBIGUOUS.
- Peptidyl transferase is a RIBOZYME.
- Stop codons: UAA, UAG, UGA.

Resources: Q11–Q16 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 56 — Thursday, 19 Nov 2026

Topic 6 — NUCLEIC ACIDS AND PROTEIN SYNTHESIS (9700 Syllabus 6.3)

Today's Sub-topic: 6.3 Gene mutations

Learning Objective: Distinguish mutation types and explain sickle cell anaemia at molecular level.

What to Learn:

- (1) Silent, Missense, Nonsense.
- (2) Frameshift: insertion/deletion.
- (3) Sickle cell: GAGGTG; GluVal; HbS polymerises.
- (4) Heterozygote advantage: HbA/HbS protection against *P. falciparum*.
- (5) Mutagens: UV, X-rays, alkylating agents.

How to Outstand:

- EXAMINER TRAP: In sickle cell, Glu Val (HYDROPHILIC HYDROPHOBIC).
- Insertion of ONE base shifts frame.
- Nonsense mutations always more severe.

Resources: Q17–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 9: 20 Nov 2026 – 26 Nov 2026

Topic: NUCLEIC ACIDS AND PROTEIN SYNTHESIS

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

DAY 57 — Friday, 20 Nov 2026

Topic 6 — NUCLEIC ACIDS AND PROTEIN SYNTHESIS (9700 Syllabus 6.1)

Today's Sub-topic: 6.1 DNA structure and replication

Learning Objective: Describe DNA double helix structure and semi-conservative replication with enzyme roles.

What to Learn:

- (1) Nucleotide: deoxyribose + phosphate + nitrogenous base.
- (2) Chargaff's rules: A=T, GC.
- (3) Double helix: antiparallel strands.
- (4) Replication enzymes: helicase, primase, DNA pol III, pol I, ligase.
- (5) Leading continuous, lagging Okazaki.
- (6) Nucleosome: DNA wound around histone octamer.

How to Outstand:

- Chargaff's calculation: if %G = 20% then %C = 20%, %A = %T = 30%.
- GC has 3 H-bonds.
- Antiparallel notation: draw 5'3' direction.
- Histones: positively charged proteins.

Resources: Q1–Q5 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 58 — Saturday, 21 Nov 2026

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 59 — Sunday, 22 Nov 2026

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 60 — Monday, 23 Nov 2026

Topic 6 — NUCLEIC ACIDS AND PROTEIN SYNTHESIS (9700 Syllabus 6.2a)

Today's Sub-topic: 6.2a Transcription

Learning Objective: Describe transcription mechanism including promoter, RNA polymerase, and eukaryotic mRNA processing.

What to Learn:

- (1) RNA pol binds promoter; synthesises mRNA 5'3'.
- (2) mRNA = same sequence as CODING strand.
- (3) Eukaryotic mRNA processing: 5' cap, poly-A tail, splicing.
- (4) RNA types: mRNA, tRNA, rRNA.
- (5) Prokaryotes: no processing; coupled transcription-translation.

How to Outstand:

- EXAMINER TRAP: mRNA has the SAME sequence as the CODING strand (with U for T).
- Template = antisense = 3'5'. Coding = sense = same as mRNA.
- Introns interrupt; exons are expressed.

Resources: Q6–Q10 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 61 — Tuesday, 24 Nov 2026

Topic 6 — NUCLEIC ACIDS AND PROTEIN SYNTHESIS (9700 Syllabus 6.2b)

Today's Sub-topic: 6.2b Translation

Learning Objective: Describe ribosome mechanism of translation including codon-anticodon pairing and peptide bond formation.

What to Learn:

- (1) Ribosome: 80S (eukaryotes), 70S (prokaryotes).
- (2) tRNA: anticodon loop, 3' CCA acceptor arm.
- (3) Translation: Initiation, Elongation, Termination.
- (4) Genetic code: triplet, non-overlapping, degenerate, universal, non-ambiguous.
- (5) Polysomes: multiple ribosomes on one mRNA.

How to Outstand:

- EXAMINER TRAP: Anticodon on tRNA is ANTIPARALLEL to mRNA codon.
- Genetic code is DEGENERATE but NON-AMBIGUOUS.
- Peptidyl transferase is a RIBOZYME.
- Stop codons: UAA, UAG, UGA.

Resources: Q11–Q16 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 62 — Wednesday, 25 Nov 2026

Topic 6 — NUCLEIC ACIDS AND PROTEIN SYNTHESIS (9700 Syllabus 6.3)

Today's Sub-topic: 6.3 Gene mutations

Learning Objective: Distinguish mutation types and explain sickle cell anaemia at molecular level.

What to Learn:

- (1) Silent, Missense, Nonsense.
- (2) Frameshift: insertion/deletion.
- (3) Sickle cell: GAGGTG; GluVal; HbS polymerises.
- (4) Heterozygote advantage: HbA/HbS protection against *P. falciparum*.
- (5) Mutagens: UV, X-rays, alkylating agents.

How to Outstand:

- EXAMINER TRAP: In sickle cell, Glu Val (HYDROPHILIC HYDROPHOBIC).
- Insertion of ONE base shifts frame.
- Nonsense mutations always more severe.

Resources: Q17–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 63 — Thursday, 26 Nov 2026

Topic 6 — NUCLEIC ACIDS AND PROTEIN SYNTHESIS (9700 Syllabus 6.1)

Today's Sub-topic: 6.1 DNA structure and replication

Learning Objective: Describe DNA double helix structure and semi-conservative replication with enzyme roles.

What to Learn:

- (1) Nucleotide: deoxyribose + phosphate + nitrogenous base.
- (2) Chargaff's rules: A=T, GC.
- (3) Double helix: antiparallel strands.
- (4) Replication enzymes: helicase, primase, DNA pol III, pol I, ligase.
- (5) Leading continuous, lagging Okazaki.
- (6) Nucleosome: DNA wound around histone octamer.

How to Outstand:

- Chargaff's calculation: if %G = 20% then %C = 20%, %A = %T = 30%.
- GC has 3 H-bonds.
- Antiparallel notation: draw 5'3' direction.
- Histones: positively charged proteins.

Resources: Q1–Q5 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 10: 27 Nov 2026 – 03 Dec 2026

Topic: NUCLEIC ACIDS AND PROTEIN SYNTHESIS

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

DAY 64 — Friday, 27 Nov 2026

Topic 6 — NUCLEIC ACIDS AND PROTEIN SYNTHESIS (9700 Syllabus 6.2a)

Today's Sub-topic: 6.2a Transcription

Learning Objective: Describe transcription mechanism including promoter, RNA polymerase, and eukaryotic mRNA processing.

What to Learn:

- (1) RNA pol binds promoter; synthesises mRNA 5'3'.
- (2) mRNA = same sequence as CODING strand.
- (3) Eukaryotic mRNA processing: 5' cap, poly-A tail, splicing.
- (4) RNA types: mRNA, tRNA, rRNA.
- (5) Prokaryotes: no processing; coupled transcription-translation.

How to Outstand:

- EXAMINER TRAP: mRNA has the SAME sequence as the CODING strand (with U for T).
- Template = antisense = 3'5'. Coding = sense = same as mRNA.
- Introns interrupt; exons are expressed.

Resources: Q6–Q10 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 65 — Saturday, 28 Nov 2026

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 66 — Sunday, 29 Nov 2026

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 67 — Monday, 30 Nov 2026

Topic 6 — NUCLEIC ACIDS AND PROTEIN SYNTHESIS (9700 Syllabus 6.2b)

Today's Sub-topic: 6.2b Translation

Learning Objective: Describe ribosome mechanism of translation including codon-anticodon pairing and peptide bond formation.

What to Learn:

- (1) Ribosome: 80S (eukaryotes), 70S (prokaryotes).
- (2) tRNA: anticodon loop, 3' CCA acceptor arm.
- (3) Translation: Initiation, Elongation, Termination.
- (4) Genetic code: triplet, non-overlapping, degenerate, universal, non-ambiguous.
- (5) Polysomes: multiple ribosomes on one mRNA.

How to Outstand:

- EXAMINER TRAP: Anticodon on tRNA is ANTIPARALLEL to mRNA codon.
- Genetic code is DEGENERATE but NON-AMBIGUOUS.
- Peptidyl transferase is a RIBOZYME.
- Stop codons: UAA, UAG, UGA.

Resources: Q11–Q16 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 68 — Tuesday, 01 Dec 2026

Topic 6 — NUCLEIC ACIDS AND PROTEIN SYNTHESIS (9700 Syllabus 6.3)

Today's Sub-topic: 6.3 Gene mutations

Learning Objective: Distinguish mutation types and explain sickle cell anaemia at molecular level.

What to Learn:

- (1) Silent, Missense, Nonsense.
- (2) Frameshift: insertion/deletion.
- (3) Sickle cell: GAGGTG; GluVal; HbS polymerises.
- (4) Heterozygote advantage: HbA/HbS protection against *P. falciparum*.
- (5) Mutagens: UV, X-rays, alkylating agents.

How to Outstand:

- EXAMINER TRAP: In sickle cell, Glu Val (HYDROPHILIC HYDROPHOBIC).
- Insertion of ONE base shifts frame.
- Nonsense mutations always more severe.

Resources: Q17–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 69 — Wednesday, 02 Dec 2026

Topic 6 — NUCLEIC ACIDS AND PROTEIN SYNTHESIS (9700 Syllabus 6.1)

Today's Sub-topic: 6.1 DNA structure and replication

Learning Objective: Describe DNA double helix structure and semi-conservative replication with enzyme roles.

What to Learn:

- (1) Nucleotide: deoxyribose + phosphate + nitrogenous base.
- (2) Chargaff's rules: A=T, GC.
- (3) Double helix: antiparallel strands.
- (4) Replication enzymes: helicase, primase, DNA pol III, pol I, ligase.
- (5) Leading continuous, lagging Okazaki.
- (6) Nucleosome: DNA wound around histone octamer.

How to Outstand:

- Chargaff's calculation: if %G = 20% then %C = 20%, %A = %T = 30%.
- GC has 3 H-bonds.
- Antiparallel notation: draw 5'3' direction.
- Histones: positively charged proteins.

Resources: Q1–Q5 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 70 — Thursday, 03 Dec 2026

Topic 6 — NUCLEIC ACIDS AND PROTEIN SYNTHESIS (9700 Syllabus 6.2a)

Today's Sub-topic: 6.2a Transcription

Learning Objective: Describe transcription mechanism including promoter, RNA polymerase, and eukaryotic mRNA processing.

What to Learn:

- (1) RNA pol binds promoter; synthesises mRNA 5'3'.
- (2) mRNA = same sequence as CODING strand.
- (3) Eukaryotic mRNA processing: 5' cap, poly-A tail, splicing.
- (4) RNA types: mRNA, tRNA, rRNA.
- (5) Prokaryotes: no processing; coupled transcription-translation.

How to Outstand:

- EXAMINER TRAP: mRNA has the SAME sequence as the CODING strand (with U for T).
- Template = antisense = 3'5'. Coding = sense = same as mRNA.
- Introns interrupt; exons are expressed.

Resources: Q6–Q10 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 11: 04 Dec 2026 – 10 Dec 2026

Topic: NUCLEIC ACIDS AND PROTEIN SYNTHESIS

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

Weekly Practice: Hamna_Bio_Topic6_NUCLEIC_ACIDS_AND_PROTEIN_SYNTHESIS.pdf
(50 Qs, 220 Marks, 3 hrs)

DAY 71 — Friday, 04 Dec 2026

Topic 6 — NUCLEIC ACIDS AND PROTEIN SYNTHESIS (9700 Syllabus 6.2b)

Today's Sub-topic: 6.2b Translation

Learning Objective: Describe ribosome mechanism of translation including codon-anticodon pairing and peptide bond formation.

What to Learn:

- (1) Ribosome: 80S (eukaryotes), 70S (prokaryotes).
- (2) tRNA: anticodon loop, 3' CCA acceptor arm.
- (3) Translation: Initiation, Elongation, Termination.
- (4) Genetic code: triplet, non-overlapping, degenerate, universal, non-ambiguous.
- (5) Polysomes: multiple ribosomes on one mRNA.

How to Outstand:

- EXAMINER TRAP: Anticodon on tRNA is ANTIPARALLEL to mRNA codon.
- Genetic code is DEGENERATE but NON-AMBIGUOUS.
- Peptidyl transferase is a RIBOZYME.
- Stop codons: UAA, UAG, UGA.

Resources: Q11–Q16 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 72 — Saturday, 05 Dec 2026

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 73 — Sunday, 06 Dec 2026

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 74 — Monday, 07 Dec 2026

Topic 7 — TRANSPORT IN PLANTS (9700 Syllabus 7.1)

Today's Sub-topic: 7.1 Xylem and phloem structure

Learning Objective: Relate structural features of xylem and phloem to their specific transport functions.

What to Learn:

- (1) Xylem vessels: dead at maturity; lignified walls; pits; no end walls.
- (2) Sieve tube elements: alive but no nucleus; sieve plates.
- (3) Companion cells: many mitochondria; actively loads sucrose.
- (4) Structural difference from other tissues.
- (5) Vascular bundle location.

How to Outstand:

- EXAMINER TRAP: Xylem requires both DEATH (hollow lumen) and LIGNIFICATION.
- Companion cells have MORE MITOCHONDRIA for ACTIVE sucrose loading.

Resources: Q1–Q4 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 75 — Tuesday, 08 Dec 2026

Topic 7 — TRANSPORT IN PLANTS (9700 Syllabus 7.2)

Today's Sub-topic: 7.2 Water transport and Casparian strip

Learning Objective: Explain apoplast/symplast pathways and the Casparian strip's role in ionic control.

What to Learn:

- (1) Apoplast: through cell walls.
- (2) Symplast: through cytoplasm.
- (3) Vacuolar pathway.
- (4) Casparian strip: band of SUBERIN in ENDODERMAL cells; blocks apoplast.
- (5) All water must enter SYMPLAST at endodermis.
- (6) Root pressure: active pumping of ions into xylem.

How to Outstand:

- EXAMINER TRAP: Casparian strip stops APOPLAST flow, forcing water into SYMPLAST.
- Suberin is HYDROPHOBIC.
- The Casparian strip is in the ENDODERMIS.

Resources: Q5–Q9 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 76 — Wednesday, 09 Dec 2026

Topic 7 — TRANSPORT IN PLANTS (9700 Syllabus 7.2b)

Today's Sub-topic: 7.2b Cohesion-tension and phloem mass flow

Learning Objective: Apply cohesion-tension theory and Münch mass flow hypothesis.

What to Learn:

- (1) Transpirational pull: generates tension.
- (2) Cohesion & Adhesion.
- (3) TACT: Transpiration + Adhesion + Cohesion + Tension.
- (4) Phloem mass flow: source to sink bulk flow.
- (5) H-ATPase proton pump + H/sucrose symporter.

How to Outstand:

- EXAMINER TRAP: Water in XYLEM moves by MASS FLOW, not osmosis.
- For mass flow in phloem: state SOURCE (high pressure), SINK (low pressure).
- TRANSLATE is phloem; TRANSPIRE is xylem.

Resources: Q10–Q15 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 77 — Thursday, 10 Dec 2026

Topic 7 — TRANSPORT IN PLANTS (9700 Syllabus 7.3)

Today's Sub-topic: 7.3 Transpiration and xerophytes

Learning Objective: Explain stomatal mechanism, potometer use, and xerophytic structural adaptations.

What to Learn:

- (1) Guard cells: K pumped IN turgid pore opens.
- (2) ABA: drought signal closes stomata.
- (3) Environmental factors: Temp, Humidity, Wind, Light.
- (4) Potometer: measures WATER UPTAKE.
- (5) Xerophytes: sunken stomata, rolled leaves, CAM.

How to Outstand:

- EXAMINER TRAP: Potometer measures WATER UPTAKE RATE, not transpiration rate.
- Guard cells have UNEQUAL wall thickening.
- ABA = abscisic acid.

Resources: Q16–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 12: 11 Dec 2026 – 17 Dec 2026

Topic: TRANSPORT IN PLANTS

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

DAY 78 — Friday, 11 Dec 2026

Topic 7 — TRANSPORT IN PLANTS (9700 Syllabus 7.1)

Today's Sub-topic: 7.1 Xylem and phloem structure

Learning Objective: Relate structural features of xylem and phloem to their specific transport functions.

What to Learn:

- (1) Xylem vessels: dead at maturity; lignified walls; pits; no end walls.
- (2) Sieve tube elements: alive but no nucleus; sieve plates.
- (3) Companion cells: many mitochondria; actively loads sucrose.
- (4) Structural difference from other tissues.
- (5) Vascular bundle location.

How to Outstand:

- EXAMINER TRAP: Xylem requires both DEATH (hollow lumen) and LIGNIFICATION.
- Companion cells have MORE MITOCHONDRIA for ACTIVE sucrose loading.

Resources: Q1–Q4 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 79 — Saturday, 12 Dec 2026

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 80 — Sunday, 13 Dec 2026

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 81 — Monday, 14 Dec 2026

Topic 7 — TRANSPORT IN PLANTS (9700 Syllabus 7.2)

Today's Sub-topic: 7.2 Water transport and Casparian strip

Learning Objective: Explain apoplast/symplast pathways and the Casparian strip's role in ionic control.

What to Learn:

- (1) Apoplast: through cell walls.
- (2) Symplast: through cytoplasm.
- (3) Vacuolar pathway.
- (4) Casparian strip: band of SUBERIN in ENDODERMAL cells; blocks apoplast.
- (5) All water must enter SYMPLAST at endodermis.
- (6) Root pressure: active pumping of ions into xylem.

How to Outstand:

- EXAMINER TRAP: Casparian strip stops APOPLAST flow, forcing water into SYMPLAST.
- Suberin is HYDROPHOBIC.
- The Casparian strip is in the ENDODERMIS.

Resources: Q5–Q9 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 82 — Tuesday, 15 Dec 2026

Topic 7 — TRANSPORT IN PLANTS (9700 Syllabus 7.2b)

Today's Sub-topic: 7.2b Cohesion-tension and phloem mass flow

Learning Objective: Apply cohesion-tension theory and Münch mass flow hypothesis.

What to Learn:

- (1) Transpirational pull: generates tension.
- (2) Cohesion & Adhesion.
- (3) TACT: Transpiration + Adhesion + Cohesion + Tension.
- (4) Phloem mass flow: source to sink bulk flow.
- (5) H-ATPase proton pump + H/sucrose symporter.

How to Outstand:

- EXAMINER TRAP: Water in XYLEM moves by MASS FLOW, not osmosis.
- For mass flow in phloem: state SOURCE (high pressure), SINK (low pressure).
- TRANSLATE is phloem; TRANSPIRE is xylem.

Resources: Q10–Q15 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 83 — Wednesday, 16 Dec 2026

Topic 7 — TRANSPORT IN PLANTS (9700 Syllabus 7.3)

Today's Sub-topic: 7.3 Transpiration and xerophytes

Learning Objective: Explain stomatal mechanism, potometer use, and xerophytic structural adaptations.

What to Learn:

- (1) Guard cells: K pumped IN turgid pore opens.
- (2) ABA: drought signal closes stomata.
- (3) Environmental factors: Temp, Humidity, Wind, Light.
- (4) Potometer: measures WATER UPTAKE.
- (5) Xerophytes: sunken stomata, rolled leaves, CAM.

How to Outstand:

- EXAMINER TRAP: Potometer measures WATER UPTAKE RATE, not transpiration rate.
- Guard cells have UNEQUAL wall thickening.
- ABA = abscisic acid.

Resources: Q16–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 84 — Thursday, 17 Dec 2026

Topic 7 — TRANSPORT IN PLANTS (9700 Syllabus 7.1)

Today's Sub-topic: 7.1 Xylem and phloem structure

Learning Objective: Relate structural features of xylem and phloem to their specific transport functions.

What to Learn:

- (1) Xylem vessels: dead at maturity; lignified walls; pits; no end walls.
- (2) Sieve tube elements: alive but no nucleus; sieve plates.
- (3) Companion cells: many mitochondria; actively loads sucrose.
- (4) Structural difference from other tissues.
- (5) Vascular bundle location.

How to Outstand:

- EXAMINER TRAP: Xylem requires both DEATH (hollow lumen) and LIGNIFICATION.
- Companion cells have MORE MITOCHONDRIA for ACTIVE sucrose loading.

Resources: Q1–Q4 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 13: 18 Dec 2026 – 24 Dec 2026

Topic: TRANSPORT IN PLANTS

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

Weekly Practice: Hamna_Bio_Topic7_TRANSPORT_IN_PLANTS.pdf (50 Qs, 220 Marks, 3 hrs)

DAY 85 — Friday, 18 Dec 2026

Topic 7 — TRANSPORT IN PLANTS (9700 Syllabus 7.2)

Today's Sub-topic: 7.2 Water transport and Casparian strip

Learning Objective: Explain apoplast/symplast pathways and the Casparian strip's role in ionic control.

What to Learn:

- (1) Apoplast: through cell walls.
- (2) Symplast: through cytoplasm.
- (3) Vacuolar pathway.
- (4) Casparian strip: band of SUBERIN in ENDODERMAL cells; blocks apoplast.
- (5) All water must enter SYMPLAST at endodermis.
- (6) Root pressure: active pumping of ions into xylem.

How to Outstand:

- EXAMINER TRAP: Casparian strip stops APOPLAST flow, forcing water into SYMPLAST.
- Suberin is HYDROPHOBIC.
- The Casparian strip is in the ENDODERMIS.

Resources: Q5–Q9 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 86 — Saturday, 19 Dec 2026

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 87 — Sunday, 20 Dec 2026

HOLIDAY LIGHT REVIEW

Flashcard/FAQ review of Topics 3–7 only, no new content.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 88 — Monday, 21 Dec 2026

HOLIDAY LIGHT REVIEW

Flashcard/FAQ review of Topics 3–7 only, no new content.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 89 — Tuesday, 22 Dec 2026

HOLIDAY LIGHT REVIEW

Flashcard/FAQ review of Topics 3–7 only, no new content.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 90 — Wednesday, 23 Dec 2026

HOLIDAY LIGHT REVIEW

Flashcard/FAQ review of Topics 3–7 only, no new content.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 91 — Thursday, 24 Dec 2026

HOLIDAY LIGHT REVIEW

Flashcard/FAQ review of Topics 3–7 only, no new content.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 14: 25 Dec 2026 – 31 Dec 2026

Topic: HOLIDAY / REVISION

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

DAY 92 — Friday, 25 Dec 2026

HOLIDAY LIGHT REVIEW

Flashcard/FAQ review of Topics 3–7 only, no new content.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 93 — Saturday, 26 Dec 2026

HOLIDAY LIGHT REVIEW

Flashcard/FAQ review of Topics 3–7 only, no new content.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 94 — Sunday, 27 Dec 2026

HOLIDAY LIGHT REVIEW

Flashcard/FAQ review of Topics 3–7 only, no new content.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 95 — Monday, 28 Dec 2026

HOLIDAY LIGHT REVIEW

Flashcard/FAQ review of Topics 3–7 only, no new content.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 96 — Tuesday, 29 Dec 2026

HOLIDAY LIGHT REVIEW

Flashcard/FAQ review of Topics 3–7 only, no new content.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 97 — Wednesday, 30 Dec 2026

HOLIDAY LIGHT REVIEW

Flashcard/FAQ review of Topics 3–7 only, no new content.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 98 — Thursday, 31 Dec 2026

HOLIDAY LIGHT REVIEW

Flashcard/FAQ review of Topics 3–7 only, no new content.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 15: 01 Jan 2027 – 07 Jan 2027

Topic: TRANSPORT IN MAMMALS

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

DAY 99 — Friday, 01 Jan 2027

HOLIDAY LIGHT REVIEW

Flashcard/FAQ review of Topics 3–7 only, no new content.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 100 — Saturday, 02 Jan 2027

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 101 — Sunday, 03 Jan 2027

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 102 — Monday, 04 Jan 2027

Topic 8 — TRANSPORT IN MAMMALS (9700 Syllabus 8.1)

Today's Sub-topic: 8.1 The heart

Learning Objective: Explain the cardiac cycle, Wiggers diagram, and conduction system.

What to Learn:

- (1) Cardiac cycle: atrial systole ventricular systole diastole.
- (2) Wiggers diagram: AV valve closure (S1), SL valve closure (S2).
- (3) Conduction: SAN AVN delay Bundle of His Purkyne fibres.
- (4) Isovolumetric contraction.
- (5) Cardiac output = $SV \times HR$.

How to Outstand:

- EXAMINER TRAP: AVN DELAYS the impulse — it does NOT initiate the heartbeat.
- Dicrotic notch = aortic semilunar valve closure.
- Purkyne fibres conduct impulse to apex ventricles contract UPWARD.

Resources: Q1–Q5 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 103 — Tuesday, 05 Jan 2027

Topic 8 — TRANSPORT IN MAMMALS (9700 Syllabus 8.2)

Today's Sub-topic: 8.2 Haemoglobin and blood transport

Learning Objective: Explain O curves, Bohr shift, fetal HbF, and CO transport.

What to Learn:

- (1) Sigmoidal ODC: positive cooperativity.
- (2) Bohr shift: CO₂ / H⁺ binds globin rightward shift.
- (3) CO transport: 85% as HCO₃⁻. Chloride shift.
- (4) HbF: gamma chains; leftward shift.
- (5) Starling's forces: HP vs OP.

How to Outstand:

- EXAMINER TRAP: CO binds globin AMINO GROUPS (not iron).
- Chloride shift maintains ELECTRICAL NEUTRALITY.
- HbF has gamma chains lower 2,3-BPG binding.

Resources: Q6–Q13 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 104 — Wednesday, 06 Jan 2027

Topic 8 — TRANSPORT IN MAMMALS (9700 Syllabus 8.3)

Today's Sub-topic: 8.3 Blood vessels and peripheral resistance

Learning Objective: Relate blood vessel layers to hemodynamic functions.

What to Learn:

- (1) Arteries: thick media; Windkessel effect.
- (2) Veins: thin media; semilunar valves.
- (3) Capillaries: single squamous cell; intercellular clefts.
- (4) Arterioles: steepest pressure drop; smooth muscle.
- (5) Double circulation advantage.

How to Outstand:

- EXAMINER TRAP: Steepest pressure drop is in ARTERIOLES.
- Windkessel effect: recoil smooths pulse.
- Capillary exchange by diffusion.

Resources: Q14–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 105 — Thursday, 07 Jan 2027

Topic 8 — TRANSPORT IN MAMMALS (9700 Syllabus 8.1)

Today's Sub-topic: 8.1 The heart

Learning Objective: Explain the cardiac cycle, Wiggers diagram, and conduction system.

What to Learn:

- (1) Cardiac cycle: atrial systole ventricular systole diastole.
- (2) Wiggers diagram: AV valve closure (S1), SL valve closure (S2).
- (3) Conduction: SAN AVN delay Bundle of His Purkyne fibres.
- (4) Isovolumetric contraction.
- (5) Cardiac output = $SV \times HR$.

How to Outstand:

- EXAMINER TRAP: AVN DELAYS the impulse — it does NOT initiate the heartbeat.
- Dicrotic notch = aortic semilunar valve closure.
- Purkyne fibres conduct impulse to apex ventricles contract UPWARD.

Resources: Q1–Q5 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 16: 08 Jan 2027 – 14 Jan 2027

Topic: TRANSPORT IN MAMMALS

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

DAY 106 — Friday, 08 Jan 2027

Topic 8 — TRANSPORT IN MAMMALS (9700 Syllabus 8.2)

Today's Sub-topic: 8.2 Haemoglobin and blood transport

Learning Objective: Explain O curves, Bohr shift, fetal HbF, and CO transport.

What to Learn:

- (1) Sigmoidal ODC: positive cooperativity.
- (2) Bohr shift: CO H binds globin rightward shift.
- (3) CO transport: 85% as HCO. Chloride shift.
- (4) HbF: gamma chains; leftward shift.
- (5) Starling's forces: HP vs OP.

How to Outstand:

- EXAMINER TRAP: CO binds globin AMINO GROUPS (not iron).
- Chloride shift maintains ELECTRICAL NEUTRALITY.
- HbF has gamma chains lower 2,3-BPG binding.

Resources: Q6–Q13 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 107 — Saturday, 09 Jan 2027

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 108 — Sunday, 10 Jan 2027

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 109 — Monday, 11 Jan 2027

Topic 8 — TRANSPORT IN MAMMALS (9700 Syllabus 8.3)

Today's Sub-topic: 8.3 Blood vessels and peripheral resistance

Learning Objective: Relate blood vessel layers to hemodynamic functions.

What to Learn:

- (1) Arteries: thick media; Windkessel effect.
- (2) Veins: thin media; semilunar valves.
- (3) Capillaries: single squamous cell; intercellular clefts.
- (4) Arterioles: steepest pressure drop; smooth muscle.
- (5) Double circulation advantage.

How to Outstand:

- EXAMINER TRAP: Steepest pressure drop is in ARTERIOLES.
- Windkessel effect: recoil smooths pulse.
- Capillary exchange by diffusion.

Resources: Q14–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 110 — Tuesday, 12 Jan 2027

Topic 8 — TRANSPORT IN MAMMALS (9700 Syllabus 8.1)

Today's Sub-topic: 8.1 The heart

Learning Objective: Explain the cardiac cycle, Wiggers diagram, and conduction system.

What to Learn:

- (1) Cardiac cycle: atrial systole ventricular systole diastole.
- (2) Wiggers diagram: AV valve closure (S1), SL valve closure (S2).
- (3) Conduction: SAN AVN delay Bundle of His Purkyne fibres.
- (4) Isovolumetric contraction.
- (5) Cardiac output = $SV \times HR$.

How to Outstand:

- EXAMINER TRAP: AVN DELAYS the impulse — it does NOT initiate the heartbeat.
- Dicrotic notch = aortic semilunar valve closure.
- Purkyne fibres conduct impulse to apex ventricles contract UPWARD.

Resources: Q1–Q5 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 111 — Wednesday, 13 Jan 2027

Topic 8 — TRANSPORT IN MAMMALS (9700 Syllabus 8.2)

Today's Sub-topic: 8.2 Haemoglobin and blood transport

Learning Objective: Explain O curves, Bohr shift, fetal HbF, and CO transport.

What to Learn:

- (1) Sigmoidal ODC: positive cooperativity.
- (2) Bohr shift: CO H binds globin rightward shift.
- (3) CO transport: 85% as HCO. Chloride shift.
- (4) HbF: gamma chains; leftward shift.
- (5) Starling's forces: HP vs OP.

How to Outstand:

- EXAMINER TRAP: CO binds globin AMINO GROUPS (not iron).
- Chloride shift maintains ELECTRICAL NEUTRALITY.
- HbF has gamma chains lower 2,3-BPG binding.

Resources: Q6–Q13 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 112 — Thursday, 14 Jan 2027

Topic 8 — TRANSPORT IN MAMMALS (9700 Syllabus 8.3)

Today's Sub-topic: 8.3 Blood vessels and peripheral resistance

Learning Objective: Relate blood vessel layers to hemodynamic functions.

What to Learn:

- (1) Arteries: thick media; Windkessel effect.
- (2) Veins: thin media; semilunar valves.
- (3) Capillaries: single squamous cell; intercellular clefts.
- (4) Arterioles: steepest pressure drop; smooth muscle.
- (5) Double circulation advantage.

How to Outstand:

- EXAMINER TRAP: Steepest pressure drop is in ARTERIOLES.
- Windkessel effect: recoil smooths pulse.
- Capillary exchange by diffusion.

Resources: Q14–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 17: 15 Jan 2027 – 21 Jan 2027

Topic: TRANSPORT IN MAMMALS

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

Weekly Practice: Hamna_Bio_Topic8_TRANSPORT_IN_MAMMALS.pdf (50 Qs, 220 Marks, 3 hrs)

DAY 113 — Friday, 15 Jan 2027

Topic 8 — TRANSPORT IN MAMMALS (9700 Syllabus 8.1)

Today's Sub-topic: 8.1 The heart

Learning Objective: Explain the cardiac cycle, Wiggers diagram, and conduction system.

What to Learn:

- (1) Cardiac cycle: atrial systole ventricular systole diastole.
- (2) Wiggers diagram: AV valve closure (S1), SL valve closure (S2).
- (3) Conduction: SAN AVN delay Bundle of His Purkyne fibres.
- (4) Isovolumetric contraction.
- (5) Cardiac output = $SV \times HR$.

How to Outstand:

- EXAMINER TRAP: AVN DELAYS the impulse — it does NOT initiate the heartbeat.
- Dicrotic notch = aortic semilunar valve closure.
- Purkyne fibres conduct impulse to apex ventricles contract UPWARD.

Resources: Q1–Q5 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 114 — Saturday, 16 Jan 2027

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 115 — Sunday, 17 Jan 2027

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 116 — Monday, 18 Jan 2027

Topic 9 — GAS EXCHANGE (9700 Syllabus 9.1)

Today's Sub-topic: 9.1 Gas exchange system histology

Learning Objective: Describe respiratory tract histology and three-layer alveolar barrier.

What to Learn:

- (1) Trachea: C-shaped cartilage, ciliated epithelium.
- (2) Bronchi: cartilage plates.
- (3) Bronchioles: NO cartilage, smooth muscle.
- (4) Alveoli: Type I (squamous), Type II (surfactant).
- (5) Three-layer diffusion barrier: $<0.5\ \mu\text{m}$.

How to Outstand:

- EXAMINER TRAP: Cartilage is ABSENT in bronchioles.
- Three layers: (1) Type I cell cytoplasm, (2) fused basement membrane, (3) capillary endothelial cell.
- Type II pneumocytes secrete SURFACTANT.

Resources: Q1–Q5 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 117 — Tuesday, 19 Jan 2027

Topic 9 — GAS EXCHANGE (9700 Syllabus 9.2)

Today's Sub-topic: 9.2 Ventilation and spirometry

Learning Objective: Explain inspiration/expiration mechanics and calculate ventilation rates.

What to Learn:

- (1) Inspiration: diaphragm & intercostals contract.
- (2) Quiet expiration: passive elastic recoil.
- (3) Forced expiration: active.
- (4) Spirometry volumes: TV, IRV, ERV, RV.
- (5) PVR and AVR calculations.

How to Outstand:

- EXAMINER TRAP: Quiet expiration is PASSIVE (elastic fibers do NOT contract).
- Residual volume CANNOT be measured by spirometry.
- $AVR = (TV - \text{dead space}) \times \text{frequency}$.

Resources: Q6–Q12 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 118 — Wednesday, 20 Jan 2027

Topic 9 — GAS EXCHANGE (9700 Syllabus 9.3)

Today's Sub-topic: 9.3 Respiratory diseases

Learning Objective: Explain pathogenesis of bronchitis, emphysema, asthma, and smoking damage.

What to Learn:

- (1) Chronic bronchitis: excess mucus, ciliostasis.
- (2) Emphysema: elastase destroys elastin.
- (3) Nicotine: vasoconstriction.
- (4) CO: binds haem carboxyhaemoglobin.
- (5) Lung cancer: carcinogens cause mutations.

How to Outstand:

- EXAMINER TRAP: Emphysema is ENZYMATIC destruction (elastase).
- CILIOSTASIS: tar paralyses cilia.
- Reid index > 0.5 = chronic bronchitis.

Resources: Q13–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 119 — Thursday, 21 Jan 2027

Topic 9 — GAS EXCHANGE (9700 Syllabus 9.1)

Today's Sub-topic: 9.1 Gas exchange system histology

Learning Objective: Describe respiratory tract histology and three-layer alveolar barrier.

What to Learn:

- (1) Trachea: C-shaped cartilage, ciliated epithelium.
- (2) Bronchi: cartilage plates.
- (3) Bronchioles: NO cartilage, smooth muscle.
- (4) Alveoli: Type I (squamous), Type II (surfactant).
- (5) Three-layer diffusion barrier: $<0.5\ \mu\text{m}$.

How to Outstand:

- EXAMINER TRAP: Cartilage is ABSENT in bronchioles.
- Three layers: (1) Type I cell cytoplasm, (2) fused basement membrane, (3) capillary endothelial cell.
- Type II pneumocytes secrete SURFACTANT.

Resources: Q1–Q5 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 18: 22 Jan 2027 – 28 Jan 2027

Topic: GAS EXCHANGE

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

Weekly Practice: Hamna_Bio_Topic9_GAS_EXCHANGE.pdf (50 Qs, 220 Marks, 3 hrs)

DAY 120 — Friday, 22 Jan 2027

Topic 9 — GAS EXCHANGE (9700 Syllabus 9.2)

Today's Sub-topic: 9.2 Ventilation and spirometry

Learning Objective: Explain inspiration/expiration mechanics and calculate ventilation rates.

What to Learn:

- (1) Inspiration: diaphragm & intercostals contract.
- (2) Quiet expiration: passive elastic recoil.
- (3) Forced expiration: active.
- (4) Spirometry volumes: TV, IRV, ERV, RV.
- (5) PVR and AVR calculations.

How to Outstand:

- EXAMINER TRAP: Quiet expiration is PASSIVE (elastic fibers do NOT contract).
- Residual volume CANNOT be measured by spirometry.
- $AVR = (TV - \text{dead space}) \times \text{frequency}$.

Resources: Q6–Q12 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 121 — Saturday, 23 Jan 2027

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 122 — Sunday, 24 Jan 2027

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 123 — Monday, 25 Jan 2027

Topic 9 — GAS EXCHANGE (9700 Syllabus 9.3)

Today's Sub-topic: 9.3 Respiratory diseases

Learning Objective: Explain pathogenesis of bronchitis, emphysema, asthma, and smoking damage.

What to Learn:

- (1) Chronic bronchitis: excess mucus, ciliostasis.
- (2) Emphysema: elastase destroys elastin.
- (3) Nicotine: vasoconstriction.
- (4) CO: binds haem carboxyhaemoglobin.
- (5) Lung cancer: carcinogens cause mutations.

How to Outstand:

- EXAMINER TRAP: Emphysema is ENZYMATIC destruction (elastase).
- CILIOSTASIS: tar paralyses cilia.
- Reid index > 0.5 = chronic bronchitis.

Resources: Q13–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 124 — Tuesday, 26 Jan 2027

Topic 10 — INFECTIOUS DISEASES (9700 Syllabus 10)

Today's Sub-topic: 10.1–10.2 Malaria and cholera

Learning Objective: Describe Plasmodium life cycle and cholerae molecular mechanism.

What to Learn:

- (1) Plasmodium life cycle: sporozoite merozoites gametocytes.
- (2) Malaria eradication challenges.
- (3) Cholera: AB toxin cAMP CFTR osmotic diarrhoea.
- (4) ORT: SGLT-1 cotransport.

How to Outstand:

- EXAMINER TRAP: Plasmodium is a PROTOCTIST.
- V. cholerae stays in INTESTINAL LUMEN.
- ORT works because SGLT-1 is UNAFFECTED by cholerae.

Resources: Q1–Q8 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 125 — Wednesday, 27 Jan 2027

Topic 10 — INFECTIOUS DISEASES (9700 Syllabus 10)

Today's Sub-topic: 10.3–10.4 Tuberculosis and HIV/AIDS

Learning Objective: Explain M. tuberculosis macrophage evasion and distinguish HIV from AIDS.

What to Learn:

- (1) TB: mycolic acid prevents phagosome-lysosome fusion.
- (2) Granuloma formation.
- (3) HIV: retrovirus, CD4+ depletion.
- (4) AIDS: CD4 < 200 or opportunistic infection.

How to Outstand:

- EXAMINER TRAP: MYCOLIC ACID prevents phagosome-LYSOSOME FUSION.
- HIV is the virus; AIDS is the syndrome.
- Opportunistic infections define AIDS.

Resources: Q9–Q15 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 126 — Thursday, 28 Jan 2027

Topic 10 — INFECTIOUS DISEASES (9700 Syllabus 10)

Today's Sub-topic: 10.5 Antibiotics and resistance

Learning Objective: Explain penicillin's mechanism and natural selection of resistance.

What to Learn:

- (1) Penicillin: blocks transpeptidase osmotic lysis.
- (2) Viruses: acellular no targets.
- (3) Resistance: spontaneous mutations selected by antibiotic.
- (4) Horizontal transfer: R-plasmids via conjugation.

How to Outstand:

- EXAMINER TRAP: Antibiotics do NOT CAUSE mutations. They are the selection pressure.
- Penicillin only kills GROWING bacteria.
- Viruses have no peptidoglycan or 70S ribosomes.

Resources: Q16–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 19: 29 Jan 2027 – 04 Feb 2027

Topic: INFECTIOUS DISEASES

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

DAY 127 — Friday, 29 Jan 2027

Topic 10 — INFECTIOUS DISEASES (9700 Syllabus 10)

Today's Sub-topic: 10.1–10.2 Malaria and cholera

Learning Objective: Describe Plasmodium life cycle and cholerae molecular mechanism.

What to Learn:

- (1) Plasmodium life cycle: sporozoite merozoites gametocytes.
- (2) Malaria eradication challenges.
- (3) Cholera: AB toxin cAMP CFTR osmotic diarrhoea.
- (4) ORT: SGLT-1 cotransport.

How to Outstand:

- EXAMINER TRAP: Plasmodium is a PROTOCTIST.
- V. cholerae stays in INTESTINAL LUMEN.
- ORT works because SGLT-1 is UNAFFECTED by cholerae.

Resources: Q1–Q8 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 128 — Saturday, 30 Jan 2027

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 129 — Sunday, 31 Jan 2027

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 130 — Monday, 01 Feb 2027

Topic 10 — INFECTIOUS DISEASES (9700 Syllabus 10)

Today's Sub-topic: 10.3–10.4 Tuberculosis and HIV/AIDS

Learning Objective: Explain M. tuberculosis macrophage evasion and distinguish HIV from AIDS.

What to Learn:

- (1) TB: mycolic acid prevents phagosome-lysosome fusion.
- (2) Granuloma formation.
- (3) HIV: retrovirus, CD4+ depletion.
- (4) AIDS: CD4 < 200 or opportunistic infection.

How to Outstand:

- EXAMINER TRAP: MYCOLIC ACID prevents phagosome-LYSOSOME FUSION.
- HIV is the virus; AIDS is the syndrome.
- Opportunistic infections define AIDS.

Resources: Q9–Q15 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 131 — Tuesday, 02 Feb 2027

Topic 10 — INFECTIOUS DISEASES (9700 Syllabus 10)

Today's Sub-topic: 10.5 Antibiotics and resistance

Learning Objective: Explain penicillin's mechanism and natural selection of resistance.

What to Learn:

- (1) Penicillin: blocks transpeptidase osmotic lysis.
- (2) Viruses: acellular no targets.
- (3) Resistance: spontaneous mutations selected by antibiotic.
- (4) Horizontal transfer: R-plasmids via conjugation.

How to Outstand:

- EXAMINER TRAP: Antibiotics do NOT CAUSE mutations. They are the selection pressure.
- Penicillin only kills GROWING bacteria.
- Viruses have no peptidoglycan or 70S ribosomes.

Resources: Q16–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 132 — Wednesday, 03 Feb 2027

Topic 10 — INFECTIOUS DISEASES (9700 Syllabus 10)

Today's Sub-topic: 10.1–10.2 Malaria and cholera

Learning Objective: Describe Plasmodium life cycle and cholerae molecular mechanism.

What to Learn:

- (1) Plasmodium life cycle: sporozoite merozoites gametocytes.
- (2) Malaria eradication challenges.
- (3) Cholera: AB toxin cAMP CFTR osmotic diarrhoea.
- (4) ORT: SGLT-1 cotransport.

How to Outstand:

- EXAMINER TRAP: Plasmodium is a PROTOCTIST.
- V. cholerae stays in INTESTINAL LUMEN.
- ORT works because SGLT-1 is UNAFFECTED by cholerae.

Resources: Q1–Q8 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 133 — Thursday, 04 Feb 2027

Topic 10 — INFECTIOUS DISEASES (9700 Syllabus 10)

Today's Sub-topic: 10.3–10.4 Tuberculosis and HIV/AIDS

Learning Objective: Explain M. tuberculosis macrophage evasion and distinguish HIV from AIDS.

What to Learn:

- (1) TB: mycolic acid prevents phagosome-lysosome fusion.
- (2) Granuloma formation.
- (3) HIV: retrovirus, CD4+ depletion.
- (4) AIDS: CD4 < 200 or opportunistic infection.

How to Outstand:

- EXAMINER TRAP: MYCOLIC ACID prevents phagosome-LYSOSOME FUSION.
- HIV is the virus; AIDS is the syndrome.
- Opportunistic infections define AIDS.

Resources: Q9–Q15 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 20: 05 Feb 2027 – 11 Feb 2027

Topic: INFECTIOUS DISEASES

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

Weekly Practice: Hamna_Bio_Topic10_INFECTIOUS_DISEASES.pdf (50 Qs, 220 Marks, 3 hrs)

DAY 134 — Friday, 05 Feb 2027

Topic 10 — INFECTIOUS DISEASES (9700 Syllabus 10)

Today's Sub-topic: 10.5 Antibiotics and resistance

Learning Objective: Explain penicillin's mechanism and natural selection of resistance.

What to Learn:

- (1) Penicillin: blocks transpeptidase osmotic lysis.
- (2) Viruses: acellular no targets.
- (3) Resistance: spontaneous mutations selected by antibiotic.
- (4) Horizontal transfer: R-plasmids via conjugation.

How to Outstand:

- EXAMINER TRAP: Antibiotics do NOT CAUSE mutations. They are the selection pressure.
- Penicillin only kills GROWING bacteria.
- Viruses have no peptidoglycan or 70S ribosomes.

Resources: Q16–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 135 — Saturday, 06 Feb 2027

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 136 — Sunday, 07 Feb 2027

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 137 — Monday, 08 Feb 2027

Topic 11 — IMMUNITY (9700 Syllabus 11.1)

Today's Sub-topic: 11.1 Phagocytosis and innate immunity

Learning Objective: Describe phagocytosis stages, antigen presentation, and non-specific defences.

What to Learn:

- (1) Non-specific defences: skin, mucus, lysozyme.
- (2) Phagocytosis: chemotaxis recognition phagosome phagolysosome.
- (3) Antigen presentation: macrophages load MHC II.
- (4) Opsonisation: antibodies coat pathogen.

How to Outstand:

- EXAMINER TRAP: The initial vesicle is called a PHAGOSOME. It only becomes a PHAGOLYSOSOME after lysosome fusion.
- MHC Class II (on APCs) vs MHC Class I (on all nucleated cells).
- PAMPs are recognised by PRRs.

Resources: Q1–Q5 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 138 — Tuesday, 09 Feb 2027

Topic 11 — IMMUNITY (9700 Syllabus 11.2)

Today's Sub-topic: 11.2 Antibodies

Learning Objective: Describe the molecular structure of antibodies and primary vs secondary responses.

What to Learn:

- (1) Immunoglobulin G structure: 4 polypeptide chains, disulfide bonds.
- (2) Antibody functions: neutralisation, agglutination, opsonisation.
- (3) Primary immune response: slow, IgM then IgG.
- (4) Secondary immune response: faster, higher IgG titre.

How to Outstand:

- EXAMINER TRAP: Antibodies do not 'kill' pathogens.
- Disulfide bonds hold heavy and light chains.
- Variable regions form antigen-binding sites.

Resources: Q6–Q10 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 139 — Wednesday, 10 Feb 2027

Topic 11 — IMMUNITY (9700 Syllabus 11.3)

Today's Sub-topic: 11.3 Cell-mediated immunity

Learning Objective: Explain roles of T-helper and cytotoxic T-cells.

What to Learn:

- (1) T-helper (CD4+): secrete cytokines.
- (2) Cytotoxic T-killer (CD8+): secrete perforin and granzymes.
- (3) T-regulatory cells: suppress immune response.

How to Outstand:

- EXAMINER TRAP: CD4+ secrete cytokines; CD8+ directly kill.
- Perforin creates pores; granzymes trigger apoptosis.

Resources: Q11–Q15 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 140 — Thursday, 11 Feb 2027

Topic 11 — IMMUNITY (9700 Syllabus 11.4)

Today's Sub-topic: 11.4 Vaccination and monoclonal antibodies

Learning Objective: Describe vaccines, herd immunity, and monoclonal antibodies production.

What to Learn:

- (1) Vaccine types.
- (2) Active vs passive immunity.
- (3) Herd immunity.
- (4) Hybridoma method: B cell + myeloma.
- (5) Applications: diagnosis and treatment.

How to Outstand:

- EXAMINER TRAP: Passive immunity is IMMEDIATE but TEMPORARY.
- Hybridoma = B cell + myeloma cell.

Resources: Q16–Q20 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

WEEK 21: 12 Feb 2027 – 15 Feb 2027

Topic: IMMUNITY

Weekly Learning Objective: Master the sub-topics allocated for this week and consolidate understanding.

Weekly Goal Checklist:

Complete daily reading

Review class notes

Complete Saturday consolidation

Optional Sunday review

Weekly Practice: Hamna_Bio_Topic11_IMMUNITY.pdf (50 Qs, 220 Marks, 3 hrs)

DAY 141 — Friday, 12 Feb 2027

Topic 11 — IMMUNITY (9700 Syllabus 11.1)

Today's Sub-topic: 11.1 Phagocytosis and innate immunity

Learning Objective: Describe phagocytosis stages, antigen presentation, and non-specific defences.

What to Learn:

- (1) Non-specific defences: skin, mucus, lysozyme.
- (2) Phagocytosis: chemotaxis recognition phagosome phagolysosome.
- (3) Antigen presentation: macrophages load MHC II.
- (4) Opsonisation: antibodies coat pathogen.

How to Outstand:

- EXAMINER TRAP: The initial vesicle is called a PHAGOSOME. It only becomes a PHAGOLYSOSOME after lysosome fusion.
- MHC Class II (on APCs) vs MHC Class I (on all nucleated cells).
- PAMPs are recognised by PRRs.

Resources: Q1–Q5 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 142 — Saturday, 13 Feb 2027

CONSOLIDATION DAY

Review the whole week, attempt 5 Qs from the pack, re-read Section D FAQs.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 143 — Sunday, 14 Feb 2027

OPTIONAL REVIEW

Flashcards / light revision, flexible.

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:

DAY 144 — Monday, 15 Feb 2027

Topic 11 — IMMUNITY (9700 Syllabus 11.2)

Today's Sub-topic: 11.2 Antibodies

Learning Objective: Describe the molecular structure of antibodies and primary vs secondary responses.

What to Learn:

- (1) Immunoglobulin G structure: 4 polypeptide chains, disulfide bonds.
- (2) Antibody functions: neutralisation, agglutination, opsonisation.
- (3) Primary immune response: slow, IgM then IgG.
- (4) Secondary immune response: faster, higher IgG titre.

How to Outstand:

- EXAMINER TRAP: Antibodies do not 'kill' pathogens.
- Disulfide bonds hold heavy and light chains.
- Variable regions form antigen-binding sites.

Resources: Q6–Q10 Section A

Self-Assessment:

Can explain this to someone else

Attempted practice questions

Reviewed mark scheme

Notes:
